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Purging Patent Law of “Private Law” Remedies

Ted Sichelman*

This Article rejects the fundamental “private law” premise of patent law remedies that courts should always attempt to make the patentee “whole” in the event of infringement because the overarching aim of patent law is to promote innovation, not to remedy private wrongs. Specifically, make-whole damages may thwart optimal innovation incentives when they concern small components of complex products involving high-switching costs, generate large consumer deadweight losses, result in substantial duplicated costs during the pre-invention R&D process, or create transaction costs far in excess of the value of the invention. In other situations, a patentee should be made more than whole. For example, inducing socially valuable innovations that do not command large profits in the private market—such as drugs for rare diseases and technologies for the disabled—may require more than make-whole compensation.

More generally, the statutory remedies provisions of the Patent Act rest on a flawed foundation. Instead of correcting for private wrongs inflicted on private parties, patent law remedies should be tailored simply to promote the types and levels of innovation that most benefit society, taking into account administrative and error costs. As such, the patent system and its associated remedies should be viewed as part of a public regulatory regime designed to further societal goals rather than a private law system that protects individual interests.

INTRODUCTION..... 518

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I. THE AIM OF PATENT LAW IS OPTIMALLY PROMOTING INNOVATION, NOT PROTECTING PRIVATE HARMS.....	529
II. RECONFIGURING PATENT LAW REMEDIES TO PROMOTE INNOVATION.....	536
A. The Standard Model of Patent Law Remedies.....	539
B. Removing Real Property: A Partially Revised Model of Patent Law Remedies.....	541
C. Fully Rejecting Tort Law Remedies in Patent Law.....	554
1. <i>Correcting Patentee Overcompensation</i>	555
2. <i>Correcting Patentee Undercompensation</i>	558
III. SOME POSSIBLE COSTS AND CRITICISMS OF JUDICIALLY MEASURED REMEDIES.....	560
A. Contravening Democracy and the Separation of Powers.....	560
B. The Costs of Uncertainty.....	562
C. Institutional and Cognitive Limitations of Courts and Judges.....	564
IV. PATENT LAW AS A PUBLIC REGULATORY MECHANISM.....	566
A. The Fundamentally Flawed Foundation of the Remedies Provisions in the Patent Act.....	566
B. <i>Towards a Regulatory Model</i> of Patent Law Remedies.....	569
CONCLUSION.....	571

Introduction

Private law—such as tort, property, and contract law—generally provides remedies for the infliction of wrongs on private individuals and entities.¹ For example, if a rambunctious bar patron punches you on the nose while you quietly sip a martini, you can sue the patron in tort for battery, collecting damages via a “liability” rule for at least the harm inflicted and—in many cases—an injunction via a “property” rule going forward to prevent future harm.² Tort law would provide you these sorts of remedies because we as a society believe the optimal state of the world is for you to go on quietly sipping your drink, free from the interference and physical damage caused by the rowdy patron. In other words, tort law—at least, traditionally—tends to assume that a world free from interference into

1. See ERNEST J. WEINRIB, *THE IDEA OF PRIVATE LAW* 143 (1995) (“When the remedy takes the form of an award of damages, a single amount undoes the injustice both of what the defendant has done and of what the plaintiff has suffered.”); Ezra Ripley Thayer, *Public Wrong and Private Action*, 27 HARV. L. REV. 317, 326–27 (1914) (discussing private civil rights of action in tort).

2. See DAN B. DOBBS, *LAW OF REMEDIES* § 8.10, at 692 (2d ed. 1993) (observing how injunctions against ongoing risky practices may accompany tort damages in personal injury cases); Guido Calabresi & A. Douglas Melamed, *Property Rules, Liability Rules, and Inalienability: One View of the Cathedral*, 85 HARV. L. REV. 1089, 1092 (1972) (proposing the property–liability rule distinction); see also 5 WILLIAM BLACKSTONE, *COMMENTARIES* 7 (arguing that private law remedies redress violations of private rights “by either restoring to [the victim] his right” or providing remuneration sufficient to compensate for the violation).

an individual's or private entity's "sphere of autonomy" is the ideal state of the world.³ As such, it generally seeks to return the private actor that has been harmed to the *status quo ante*, through damages, an injunction, or both.⁴

Patent law, on the other hand, is not designed to remedy private wrongs. Rather, its major aim is to promote innovation.⁵ Nonetheless, patent remedies mirror traditional tort law remedies by attempting to restore the patentee to the *status quo ante*—namely, the state of the world in which there is no infringement of the patent.⁶ No infringement occurs when third parties, including competitors, do not practice the patent or pay license fees to permissibly perform otherwise infringing acts.⁷ Historically, patent law

3. See Donald P. Judges, *Of Rocks and Hard Places: The Value of Risk Choice*, 42 EMORY L.J. 1, 63 (1993) ("Tort law generally seeks to protect the autonomy of plaintiffs by compensating for and deterring the forced intrusions of defendants' tortious conduct."). Of course, there are those exceptional cases for which intentional intrusions into a person's sphere of autonomy are considered excused or justified, such as self-defense, duress, and the like. See, e.g., RESTATEMENT (SECOND) OF TORTS § 65 (1965) (setting forth factors for establishing self-defense).

4. See Aaron Xavier Fellmeth, *Civil and Criminal Sanctions in the Constitution and Courts*, 94 GEO. L.J. 1, 60 (2005) ("In tort law, injunctions are granted . . . when the remedy of compensatory damages will not suffice to restore the status quo ante."); Karen E. Sandrik, *Reframing Patent Remedies*, 67 U. MIAMI L. REV. 95, 102 (2012) ("Further, patent and tort remedies largely mirror one another in that the goal of both remedial structures is to restore the aggrieved party to the status quo ante." (citing an earlier version of this Article)); see also RESTATEMENT (SECOND) OF TORTS § 936 (1979) (setting forth factors for when injunctions are appropriate).

5. See U.S. CONST. art. I, § 8, cl. 8 ("To promote the Progress of Science and useful Arts . . .") (emphasis added); *Sears, Roebuck & Co. v. Stiffel Co.*, 376 U.S. 225, 229–31 (1964) ("Patents are not given as favors, as was the case of monopolies given by the Tudor monarchs, . . . but are meant to encourage invention by rewarding the inventor with the right . . . to exclude others from the use of his invention."); Peter S. Menell, *Intellectual Property: General Theories*, in 2 ENCYCLOPEDIA OF LAW AND ECONOMICS 129, 130–33 (Boudewijn Bouckaert & Gerrit De Geest eds., 2000) ("The United States Constitution expressly conditions the grant of power to Congress to create patent and copyright laws upon a utilitarian foundation . . ."). By using the term *innovation*, I intend to include not only invention but the commercialization and dissemination of the invention. See, e.g., Jan Fagerberg, *Innovation: A Guide to the Literature*, in THE OXFORD HANDBOOK OF INNOVATION 1, 4 (Jan Fagerberg et al. eds., 2005) ("Invention is the first occurrence of an idea for a new product or process, while innovation is the first attempt to carry it out into practice.").

6. Mark A. Lemley, *Distinguishing Lost Profits From Reasonable Royalties*, 51 WM. & MARY L. REV. 655, 674 (2009) ("Patent damages are supposed to compensate patent owners for their losses, putting them back in the world they would have inhabited but for infringement."); see also Thomas F. Cotter, *Four Principles for Calculating Reasonable Royalties in Patent Infringement Litigation*, 27 SANTA CLARA COMPUTER & HIGH TECH. L.J. 725, 727 (2011) ("[T]he baseline damages recovery for prevailing patent owners should be the amount that restores them to the position they would have enjoyed but for the infringement.").

7. See 35 U.S.C. § 271(a) (2006) ("[W]hoever without authority makes, uses, offers to sell, or sells any patented invention, within the United States or imports into the United States any patented invention during the term of the patent therefor, infringes the patent."). There is also a fairly limited "experimental use" exception to patent infringement, which excuses otherwise

typically provided for damages for past infringement and injunctions on a going-forward basis to restore the *status quo ante*.⁸ Backward-looking damages compensate the patentholder either for lost profits caused by the third party's infringing activity or for forgone royalties in the event the patentholder did not practice its patent (or, in some situations, where royalties would result in greater damages than lost profits).⁹ Forward-looking injunctions restore the patentholder to the equitable *status quo ante* at the time of patent issuance, providing the holder an absolute right—backed by contempt sanctions—to prevent third parties from infringing the patent.¹⁰

In *eBay v. MercExchange*,¹¹ the Supreme Court shifted the calculus by holding that injunctions are not mandatory and should instead be awarded on the basis of a set of equitable factors.¹² In an influential concurrence,¹³

infringing acts. See 5 DONALD S. CHISUM, CHISUM ON PATENTS § 16.03[1] (2012) (collecting cases addressing the experimental use doctrine).

8. See, e.g., *Richardson v. Suzuki Motor Co.*, 868 F.2d 1226, 1247 (Fed. Cir. 1989) (reaffirming the long-standing rule of issuing injunctions to successful patentees); *Connell v. Sears, Roebuck & Co.*, 722 F.2d 1542, 1548 (Fed. Cir. 1983) (“[A] patent is a form of property right, and the right to exclude recognized in a patent is but the essence of the concept of property.”); see also *Smith Int'l, Inc. v. Hughes Tool Co.*, 718 F.2d 1573, 1577–78 (Fed. Cir. 1983) (reasoning that the incentive to engage in research would be diminished without the right to injunctive relief); Ben Depoorter, *Property Rules, Liability Rules and Patent Market Failure*, 4 ERASMUS L. REV. 59, 61 (2008) (“[T]he equitable remedy of injunction has dominated the law of intellectual property.”).

9. See 35 U.S.C. § 284 (2006 & Supp. V 2012) (providing for “no . . . less than a reasonable royalty for the use made of the invention by the infringer”); Lemley, *supra* note 6, at 655 (“Patent damages are designed to compensate patentees for their losses, not punish accused infringers or require them to disgorge their profits.”); *id.* at 657 (“Giving patentees the profits they would have made absent the infringement effectively puts them in the same position as if they had had an injunction in place all along.”); see also ROGER D. BLAIR & THOMAS F. COTTER, *INTELLECTUAL PROPERTY: ECONOMIC AND LEGAL DIMENSIONS OF RIGHTS AND REMEDIES* 12 (2005) (noting that damages for an infringement “may include an award of the plaintiff’s lost profits attributable to the infringement; the amount of an established royalty; or a reasonable royalty”).

10. See *Osborn v. Bank of U.S.*, 22 U.S. (9 Wheat.) 738, 749 (1824) (stating that patent owners may obtain injunctions to prevent others from using the patent based on “the principle[] that the injury was consequential, not direct, and that it would be difficult, if not impossible, to estimate the damages”); *Rite Hite Corp. v. Kelley Co.*, 56 F.3d 1538, 1562 (Fed. Cir. 1995) (“An injunction preserves the patentee’s exclusive right to market embodiments of the patented invention.”).

11. 547 U.S. 388 (2006).

12. See *id.* at 391–92 (holding that the well-established principles of equitable relief “apply with equal force to disputes arising under the Patent Act”); Carl Shapiro, *Injunctions, Hold-Up, and Patent Royalties*, 12 AM. L. & ECON. REV. 280, 282 (2010) (“The Supreme Court ruled unanimously that the district court has discretion whether to grant or deny injunctive relief based on traditional principles of equity, using a four-factor test.”).

13. See *eBay*, 547 U.S. at 395–97 (Kennedy, J., concurring). A large number of opinions have specifically cited Justice Kennedy’s concurring opinion with approval. See, e.g., *Robert Bosch LLC v. Pylon Mfg. Corp.*, 659 F.3d 1142, 1150 (Fed. Cir. 2011); *Salinger v. Colting*, 607 F.3d 68, 82 (2d Cir. 2010); *N. Am. Med. Corp. v. Axiom Worldwide, Inc.*, 522 F.3d 1211, 1228 (11th Cir. 2008); *ATCS Int’l LLC v. Jefferson Contracting Corp.*, 807 F. Supp. 2d 516, 519 (E.D. Va. 2011); *i4i Ltd. P’ship v. Microsoft Corp.*, 670 F. Supp. 2d 568, 600 (E.D. Tex. 2009); *Hynix*

Justice Kennedy argued that entities that do not practice their patents (so-called nonpracticing entities or “NPEs”)—typically, by forgoing manufacturing and product sales in favor of licensing—generally should not be awarded an injunction because it would give them “undue leverage” over third parties, particularly when the patent covers a “small component” of a complex product.¹⁴ Implicit in Justice Kennedy’s reasoning is the tort law *status quo ante* rationale: because an NPE would by definition license its patents, a liability rule providing damages on a forward-looking basis—assuming it adequately reflected the market royalty rate—would return the NPE exactly to that state of the world that would have existed but for the infringement.¹⁵ Conversely, Justice Kennedy assumed that a practicing entity would not generally license its patents, instead choosing to leverage its patents by earning supernormal profits from selling products and

Semiconductor Inc. v. Rambus Inc., 609 F. Supp. 2d 951, 966 (N.D. Cal. 2009); Amgen, Inc. v. F. Hoffman-La Roche Ltd., 581 F. Supp. 2d 160, 211 (D. Mass. 2008); MGM Studios, Inc. v. Grokster, Ltd., 518 F. Supp. 2d 1197, 1215–16 (C.D. Cal. 2007); Commonwealth Sci. and Indus. Research Org. v. Buffalo Tech. Inc., 492 F. Supp. 2d 600, 605 (E.D. Tex. 2007); MPT, Inc. v. Marathon Labels, Inc., 505 F. Supp. 2d 401, 419–20 (N.D. Ohio 2007); z4 Techs., Inc. v. Microsoft Corp., 434 F. Supp. 2d 437, 441 (E.D. Tex. 2006). Additionally, a well-known empirical study indicates that Justice Kennedy’s opinion has been far more influential than Chief Justice Roberts’s opinion, which suggested following the historical practice of typically awarding injunctive relief. See *eBay*, 547 U.S. at 394–95 (Roberts, C.J., concurring) (“From at least the early 19th century, courts have granted injunctive relief upon a finding of infringement in the vast majority of patent cases. . . . When it comes to discerning and applying those standards, in this area as others, a page of history is worth a volume of logic.” (internal quotation marks omitted)); FTC, THE EVOLVING IP MARKETPLACE: ALIGNING PATENT NOTICE AND REMEDIES WITH COMPETITION app. b at 256–59 (2001), available at <http://www.ftc.gov/sites/default/files/documents/reports/evolving-ip-marketplace-aligning-patent-notice-and-remedies-competition-report-federal-trade/110307patentreport.pdf> (presenting the results of Steven Malin’s survey of post-*eBay* cases, which found that district courts have denied injunctions to nonpracticing entities more than 50% of the time).

14. See *eBay*, 547 U.S. at 396–97 (Kennedy, J., concurring) (“When the patented invention is but a small component of the product . . . and the threat of an injunction is employed simply for undue leverage in negotiations, legal damages may well be sufficient to compensate for the infringement and an injunction may not serve the public interest.”).

15. See *id.* at 396 (“For [NPEs], an injunction, and the potentially serious sanctions arising from its violation, can be employed as a bargaining tool to charge exorbitant fees to companies that seek to buy licenses to practice the patent.”). Presumably, Justice Kennedy viewed the fees generated from an injunction as “exorbitant,” see *id.*, for NPEs, but not for practicing entities, because—by widely licensing instead of practicing (or exclusively licensing) their patents—NPEs do not seek to exclude others from practicing their patents in the marketplace, nor do NPEs market products potentially subject to an accused infringer’s patents. See FTC, TO PROMOTE INNOVATION: THE PROPER BALANCE OF COMPETITION AND PATENT LAW AND POLICY, ch. 3, at 38–39 (2003), available at <http://www.ftc.gov/os/2003/10/innovationrpt.pdf> (“The potential for hold-up to result in mutually assured destruction means firms actively participating in the industry—patent practicing entities (PPEs)—are unlikely to employ this hold-up strategy against each other. . . . [H]owever, identified firms referred to as non-practicing entities (NPEs) . . . can successfully employ a hold-up strategy without fear of retaliation.”) cited in *eBay*, 547 U.S. at 396 (Kennedy, J., concurring). As such, any leverage gained by NPEs from the threat of curtailing such activity via an injunction as litigation remedy is “undue” on this view. *eBay*, 547 U.S. at 396–97 (Kennedy, J., concurring).

services in the marketplace.¹⁶ Importantly, both the historical doctrine and slightly modified approach of *eBay*¹⁷ assume that the *status quo ante* endgame, to the extent it can be costlessly and accurately implemented by a court, is in fact the ideal remedy.¹⁸ In other words, the courts—and essentially the entire body of academic literature—have assumed that treating the patentee like a private right holder entitled to traditional private law remedies optimally promotes innovation.¹⁹

I argue in this Article, however, that the traditional view is wrong in at least three contexts. First, along the lines of Justice Kennedy’s suggestion, when a patent covers a minor component of a complex product that a third party has already implemented—and there are large switching costs in implementing a substitute for the patented component—then providing any

16. See *eBay*, 547 U.S. at 396 (Kennedy, J., concurring) (“An industry has developed in which firms use patents not as a basis for producing and selling goods but, instead, primarily for obtaining licensing fees.”).

17. Justice Thomas’s and Chief Justice Roberts’s opinions in *eBay* make no explicit distinction between practicing entities and NPEs. See *id.* at 390–95 (majority opinion and Roberts, J., concurring). Following *eBay*, however, lower courts have followed Justice Kennedy’s distinction—routinely granting injunctions to practicing entities but denying them to NPEs. See FTC, *supra* note 13 (presenting findings that, in cases in which practicing and nonpracticing entities were distinguished, district courts granted injunctions to practicing entities about 85% of the time but only about 45% of the time to NPEs following *eBay*).

18. See *Aro Mfg. Co. v. Convertible Top Replacement Co.*, 377 U.S. 476, 507 (1964) (holding that in determining damages, the question is “had the Infringer not infringed, what would Patent Holder-Licensee have made?” (internal quotation marks omitted)); *Pall Corp. v. Micron Separations, Inc.*, 66 F.3d 1211, 1223 (Fed. Cir. 1995) (“[T]he purpose of compensatory damages is not to punish the infringer, but to make the patentee whole.”); see also Amy L. Landers, *Patent Valuation Theory and the Economics of Improvement*, 88 TEXAS L. REV. SEE ALSO 163, 166 (2009) (“[P]atent damages are a make-whole remedy, intended to restore the patentee to the same position as before infringement.”).

19. See *General Motors Corp. v. Devex Corp.*, 461 U.S. 648, 653–55 (1983) (“When Congress wished to limit an element of recovery in a patent infringement action, it said so explicitly. . . . Congress’ overriding purpose [is] affording patent owners complete compensation.”); 6 JOHN GLADSTONE MILLS III ET AL., *PATENT LAW FUNDAMENTALS* § 20:65 (2d ed. 2013) (“[I]t is only by means of injunctive relief that a patentee can realize ‘the right to exclude others’”); Roger D. Blair & Thomas F. Cotter, *Rethinking Patent Damages*, 10 TEX. INTELL. PROP. L.J. 1, 4 (2001) (“[W]e recommend the application of traditional tort-law doctrines of cause-in-fact and proximate cause to patent questions, and generally reject the idea that patent infringement is materially different”); Frank H. Easterbrook, *Intellectual Property Is Still Property*, 13 HARV. J.L. & PUB. POL’Y 108, 109 (1990) (“Patents give a right to exclude, just as the law of trespass does with real property.”); Paul J. Heald, *Optimal Remedies for Patent Infringement: A Transactional Model*, 45 HOUS. L. REV. 1165, 1171 (2008) (“In establishing what constitutes infringement and what remedies apply, patent law’s secondary function looks like tort law.”); Amy L. Landers, *Liquid Patents*, 84 DENV. U. L. REV. 199, 252–53 (2006) (contending that patent remedies should differ as between practicing and nonpracticing entities because each incurs different kinds of “harm”); Mark A. Lemley & Carl Shapiro, *Patent Holdup and Royalty Stacking*, 85 TEXAS L. REV. 1991, 2036 (2007) (arguing that retaining injunctions for practicing entities “is justified in part for reasons of equity”); see also Megan M. La Belle, *Patent Law as Public Law*, 20 GEO. MASON L. REV. 41, 41 (2012) (“Patent litigation historically has been regarded as private law litigation, meaning disputes between private parties about private rights.” (internal quotation marks omitted)).

patentee, *practicing or not*, with an injunction on a going-forward basis will generally yield market rewards (or settlement payments) far in excess of the value of the innovative component to society.²⁰ This windfall to the patentee results in the patent system providing too great an incentive for component and incremental innovations relative to discrete, whole product innovations.²¹ Indeed, Mark Lemley and Carl Shapiro have convincingly argued as much in the context of nonpracticing entities.²² Yet, because Lemley, Shapiro, and Justice Kennedy fall prey to the assumption that patent remedies should mirror traditional private law remedies, they mistakenly conclude that injunctions should still be generally available to practicing entities.²³ Rather, as I show below, their arguments against issuing injunctions to NPEs are often just as applicable to practicing entities.²⁴

Second, in many contexts, the patent system provides excessive incentives to generate needed R&D and commercialization activity.²⁵ For example, the costs of invention and commercialization in the software industry are far below those in the pharmaceutical industry.²⁶ Yet, the

20. See *infra* subpart II(B).

21. For instance, such differential incentives may unduly increase incentives to innovate—or at least seek patent protection for plausible innovations—in the software industry relative to the pharmaceutical industry. See Julie E. Cohen & Mark A. Lemley, *Patent Scope and Innovation in the Software Industry*, 89 CALIF. L. REV. 1, 26 n.94 (2001) (noting that a patented invention typically covers a small part of a software product as compared with an entire pharmaceutical product). Relatedly, it may provide an inefficient advantage to large firms, which tend to incrementally innovate, relative to small firms, which tend to radically innovate. See Jonathan M. Barnett, *Is Intellectual Property Trivial?*, 157 U. PA. L. REV. 1691, 1736–37 (2009) (“[L]arge firms tend to undertake low-risk, incremental innovation projects that preserve market share while small firms tend to undertake high-risk, radical innovation projects that seek to capture market share.”).

22. See generally Lemley & Shapiro, *supra* note 19 (discussing some of the problems associated with providing injunctive relief to patentees). As an important point of clarification, Lemley would effectively treat an NPE that exclusively licenses a patent as a practicing entity, since the NPE essentially stands in the shoes of the sole practicing entity from a market perspective. Lemley, *supra* note 6, at 673.

23. See *infra* subparts II(B)–(C).

24. See *infra* subparts II(B)–(C).

25. See Mark A. Lemley, *Property, Intellectual Property, and Free Riding*, 83 TEXAS L. REV. 1031, 1058–65 (2005) (positing that providing patents with the full social value of an invention would often result in overcompensation); see also Vincenzo Denicolò, *Do Patents Over-Compensate Innovators?*, 22 ECON. POL’Y 679, 713 (2007) (proposing a model to determine whether patent owners are overcompensated by comparing profit ratios to the elasticity of the supply of inventions); cf. Peter S. Menell, *The Challenges of Reforming Intellectual Property Protection for Computer Software*, 94 COLUM. L. REV. 2644, 2646 (1994) (“Excessive protection for first generation innovation can impede later stages, thereby undermining some of the salutary effects of strong intellectual property protection.”).

26. See Dan L. Burk & Mark A. Lemley, *Policy Levers in Patent Law*, 89 VA. L. REV. 1575, 1581–82 (2003) (noting that R&D costs for software are considerably lower than those for pharmaceuticals); Gregory N. Mandel, *Will America Reinvent Itself? Patent Reform in 2011*, BUS. L. TODAY, Aug. 2011, at 1, 2 (“Developing a new drug or biologic routinely takes a decade or

duration of software and pharmaceutical patents are exactly the same (indeed, in practice, software patents last longer),²⁷ and—at least in rough conceptual terms—the scope of software patents often exceeds the scope of pharmaceutical patents.²⁸ If the broad scope and long duration afforded software patents is unnecessary to incentivize innovation in that industry, then the rewards provided by the patent system are excessive.²⁹ This result can create windfalls for innovators, which in turn can foster needless consumer deadweight losses, particularly when (1) the patentholder enjoys the ability to price its patented goods over the competitive price,³⁰ or

more, costs hundreds of millions or billions of dollars, and often requires testing hundreds of alternatives or compounds. . . . New software applications can be produced on much shorter time scales and for a much more limited investment”); *see also* PETER TOLLMAN ET AL., THE BOST. CONSULTING GRP., A REVOLUTION IN R&D: HOW GENOMICS AND GENETICS ARE TRANSFORMING THE BIOPHARMACEUTICAL INDUSTRY 12 (2001) (estimating that the cost to discover, develop, and commercialize each patented drug is about \$880 million); Shanling Li et al., *Why Do Software Firms Fail? Capabilities, Competitive Actions, and Firm Survival in the Software Industry from 1995 to 2007*, 21 INFO. SYS. RES. 631, 642, 643 tbl.1 (2010) (surveying 870 publicly owned software companies and finding that average R&D expenditures per company were approximately \$27 million annually).

27. *See* Emily Michiko Morris, *The Myth of Generic Pharmaceutical Competition Under the Hatch-Waxman Act*, 22 FORDHAM INTELL. PROP. MEDIA & ENT. L.J. 245, 266–67 (2012) (noting that the effective patent term for pharmaceuticals, even with patent term restoration under the Hatch-Waxman Act, is fourteen years while the effective term for nonpharmaceutical patents is eighteen-and-a-half years); *see also* Daniel R. Cahoy, *An Incrementalist Approach to Patent Reform Policy*, 9 N.Y.U. J. LEGIS. & PUB. POL’Y 587, 648 (2006) (arguing that the effective patent term for software patents is too long and for pharmaceutical patents is too short).

28. *See* Emily Michiko Morris, *Res or Rules? Patents and the (Uncertain) Rules of the Game*, 18 MICH. TELECOMM. & TECH. L. REV. 481, 498–99 (2012) (stating that the scope of software patents is defined more by concepts than by physical or functional structures, unlike chemical patents, whose scope corresponds only to a limited number of chemical structures); *see also* Robert P. Merges & Richard R. Nelson, *On the Complex Economics of Patent Scope*, 90 COLUM. L. REV. 839, 843 (1990) (explaining that effective patent scope depends on the type of technology at issue).

29. *See* Burk & Lemley, *supra* note 26, at 1687–88 (“While most biotechnological and chemical inventions require broad patent protection because of their high cost and uncertain development process, the opposite is true in the case of software development. Software inventions tend to have a quick, cheap, and fairly straightforward post-invention development cycle.”); Linda R. Cohen & Roger G. Noll, *Intellectual Property, Antitrust and the New Economy*, 62 U. PITT. L. REV. 453, 469 (2001) (arguing that software patents provide excessive protection); Richard R. Nelson, *Intellectual Property Protection for Cumulative Systems Technology*, 94 COLUM. L. REV. 2674, 2674 (1994) (arguing for a moderate protection scheme to meet the protective needs of the software industry).

30. WILLIAM D. NORDHAUS, *INVENTION, GROWTH, AND WELFARE: A THEORETICAL TREATMENT OF TECHNOLOGICAL CHANGE* 76 (1969) (describing a model for determining an optimal patent term by balancing increased incentives for innovation against greater deadweight losses); Jonathan M. Barnett, *Private Protection of Patentable Goods*, 25 CARDOZO L. REV. 1251, 1269 (2004) (noting that a patent’s social costs include “supracompetitive pricing power exerted by the patent holder (or, more specifically, the deadweight loss resulting from the patent holder’s output restrictions)”).

(2) when multiple parties needlessly duplicate R&D in “racing” for an excessive patent prize.³¹

Third, when reasonable minds differ over whether a given patent is infringed, valid, or enforceable, it may be economically efficient for third parties to forgo large transaction costs in negotiating a license, choosing instead to infringe.³² Like the theory of efficient breach in contract law,³³ I argue below that efficient infringement can occur when the transaction costs of negotiation dwarf the value of the innovation at issue, which can result when there is large uncertainty in the underlying patent rights or simply when the economic value of the innovation is fairly minimal.³⁴ In these situations, it may be optimal to preclude injunctions in order to foster efficient infringement.³⁵

On the other hand, I assert that while full compensatory damages may be excessive in the three situations described above, there are at least two other situations in which the patentee may need *more than* compensatory damages to generate optimal innovation incentives.³⁶ First, if detection of infringement is difficult, damages should be enhanced to compensate for undetected infringement—activity for which the accused infringer reaps profit but pays no reward to the patentee.³⁷ This condition should apply

31. See SUZANNE SCOTCHMER, *INNOVATIONS AND INCENTIVES* 112–13 (2004) (discussing how broad intellectual property rights can incite “races” for patents, resulting in duplicated costs from different inventors expending time and money to achieve the same goal inefficiently); Partha Dasgupta & Joseph Stiglitz, *Uncertainty, Industrial Structure, and the Speed of R&D*, 11 BELL J. ECON. 1, 11–14 (1980) (describing how parties engage in duplicative R&D in a monopolistic controlled market in a race to attain the monopoly pricing prize); Mark F. Grady & Jay I. Alexander, *Patent Law and Rent Dissipation*, 78 VA. L. REV. 305, 308 (1992) (“The defect in the system is that if multiple inventors expend resources in competition for the patent monopoly, the benefit to society of having the invention will be dissipated by the cost of numerous, redundant, development efforts.”); see also Kenneth W. Dam, *The Economic Underpinnings of Patent Law*, 23 J. LEGAL STUD. 247, 252 n.14 (1994) (“A further aspect of rent seeking beyond unnecessary duplication of R&D expenditures is that the race for the patent will cause R&D expenditures to be made at a faster than optimal rate.”).

32. See *infra* section II(C)(1).

33. See Ian R. Macneil, *Efficient Breach of Contract: Circles in the Sky*, 68 VA. L. REV. 947, 950–53 (1982) (positing that “efficient breach” is efficient when the transaction costs of renegotiating the contract outweigh the transaction costs from breach).

34. See *infra* section II(C)(1).

35. Cf. Ted Sichelman, *Commercializing Patents*, 62 STAN. L. REV. 341, 345–47 (2010) (proposing a new “commercialization” patent that would grant an affirmative equitable right to make and sell a product that could infringe a traditional “invention” patent).

36. See *infra* section II(C)(2).

37. See Michael Abramowicz, *A Unified Economic Theory of Noninfringement Opinions*, 14 FED. CIR. B.J. 241, 254 (2004) (“If infringers pay full damages, their conduct will be optimized, so if patentees will enforce their rights only some of the time, enhanced damages are appropriate, with the enhanced damages multiplier equal to the inverse of the probability of detection.”); Roger D. Blair & Thomas F. Cotter, *An Economic Analysis of Damages Rules in Intellectual Property Law*, 39 WM. & MARY L. REV. 1585, 1591 (1998) (remarking that damage multipliers may be necessary to compensate for low detection levels).

regardless of whether the infringer was “willful,” unless the willfulness somehow increases the odds of nondetection. Second, if the private economic value of a patented innovation to a single patentee is substantially less than the social value of the innovation—so much so that the patentee has insufficient incentives to invest in R&D and commercialization to produce the innovation—then the patentee will need more reward than mere compensatory damages in the event of infringement.³⁸ However, this reward multiplier should not typically be borne by the infringer—who, like the patentee, generally enjoys none of the greater social benefits of the innovation—but rather by society as a whole (for example, the government).³⁹

More generally, I argue that instead of focusing on the substantive rule at issue—here, identifying those actions that should count as “patent infringement”—policymakers and scholars should also examine ways to adjust the manner of enforcement, judicial procedure, and remedies to achieve effective substantive aims.⁴⁰ When the cost of particularized substantive rulemaking is high, as in the case of patent law,⁴¹ particularized enforcement, procedure, and remedies may provide a better route for achieving optimal outcomes.⁴² Unfortunately, modifying enforcement

38. As I argue below, most patentees will not require the full social value of their invention to appropriately incentivize them. See *infra* subpart II(B). However, there will surely be situations for which that is not so. Cf. Suzanne Scotchmer, *Standing on the Shoulders of Giants: Cumulative Research and the Patent Law*, J. ECON. PERSP., Winter 1991, at 29, 31 (arguing that the only way to ensure that socially desirable innovations are researched is to allow research firms to “collect as revenue all the social value they create”); Steven Shavell & Tanguy van Ypersele, *Rewards Versus Intellectual Property Rights*, 44 J.L. & ECON. 525, 529 (2001) (“[I]ncentives to invest in research are inadequate because monopoly profits are less than the social surplus created by an innovation.”).

39. See *infra* section II(C)(2).

40. See *infra* subpart II(C).

41. Patent law is rife with industry variation in the economics of innovation incentives, but Congress and the courts have had difficulty in substantially differentiating the law for specific industries. Alan Devlin, *Patent Law’s Parsimony Principle*, 25 BERKELEY TECH. L.J. 1693, 1707–08 (2010) (“[B]ecause it is so difficult to determine why particular inventors innovate, patent law has typically declined to incorporate inventor- or even industry-specific principles into its doctrine[, generally operating instead] on a ‘one-size-fits-all’ basis, attempting to spur optimal levels of innovation through the provision of largely uniform reward structures.” (footnotes omitted)). Burk and Lemley argue that patent law is effectively differentiated in application. See Dan L. Burk & Mark A. Lemley, *Is Patent Law Technology-Specific?*, 17 BERKELEY TECH. L.J. 1155, 1156–57 (2002) (discussing the fact that general legal standards lead to diverse results in different technology areas such as software and pharmaceuticals); Burk & Lemley, *supra* note 26, at 1577 (“[D]espite the appearance of uniformity, patent law is actually as varied as the industries it seeks to foster.”). However, such differentiation is ultimately constrained by patent law’s textual uniformity and cannot achieve fully efficient differentiation. See Clarisa Long, *Our Uniform Patent System*, FED. LAW., Feb. 2008, at 44, 48 (“At present, the patent statute does provide for some technology-specific variation and exempts certain groups from liability, but these are the rare exceptions rather than the rule.”).

42. Cf. Leandra Lederman & Ted Sichelman, *Enforcement as Substance in Tax Compliance*, 70 WASH. & LEE L. REV. 1679, 1747–49 (2013) (contending that tailoring enforcement rates by

approaches has generally been overlooked as a means for compensating for defects in the primary substantive law at issue.⁴³ Most analytical treatments of enforcement emphasize its costly nature and concern various mechanisms for increasing compliance while maintaining or reducing administrative costs.⁴⁴ In these models, if enforcement were costless, the ideal approach would be to punish legal actors for every violation of the law in order to ensure 100% compliance.⁴⁵

However, Ian Ayres and Paul Klemperer have offered an alternative model,⁴⁶ extended by me in other work,⁴⁷ in which *the aim of enforcement is less than 100% compliance*.⁴⁸ Specifically, we show that imperfect, probabilistic enforcement of a patent may result in welfare outcomes superior to ironclad enforcement.⁴⁹ Such a counterintuitive goal is justified when the applicable substantive rule generates substantial unnecessary costs—for example, deadweight losses and duplicated development—and these costs cannot be easily remedied at the legislative level.⁵⁰ In these instances, modifying enforcement in the judicial or executive domains may significantly reduce the costs imposed by the substantive rule.⁵¹ If these modification costs are relatively low, then “measuring” enforcement or

industry and product markets can sometimes reduce deadweight losses and other social costs resulting from taxation more efficiently than modifying the substantive tax law).

43. See *id.* at 1681 (“Scholars often assume that perfect enforcement of the laws, though unrealistic, is the ideal, and have focused on achieving the highest level of compliance possible at the lowest cost.”).

44. *Id.*; cf. Gary S. Becker & George J. Stigler, *Law Enforcement, Malfeasance, and Compensation of Enforcers*, 3 J. LEG. STUD. 1, 1 (1974) (“Both the normative and positive approaches to legislation . . . generally have taken enforcement of laws for granted, and have not included systematic analyses of the cost of enforcing different kinds of laws.”).

45. Lederman & Sichelman, *supra* note 42, at 1690–93; see also Blair & Cotter, *supra* note 37, at 1619 (“In order to deter infringement, we must have a set of rules that renders an infringement unprofitable.”); Blair & Cotter, *supra* note 19, at 9 (assuming that infringement will necessarily reduce “incentive[s] to innovate” below the optimal level).

46. Ian Ayres & Paul Klemperer, *Limiting Patentees’ Market Power Without Reducing Innovation Incentives: The Perverse Benefits of Uncertainty and Non-Injunctive Remedies*, 97 MICH. L. REV. 985, 988–89 (1999) (“[T]his Article shows that a regime with some uncertainty and delay can produce this reward more efficiently than a regime in which enforcement is instantaneous and certain.”).

47. See Ted Sichelman, *Quantum Game Theory and Coordination in Intellectual Property* 5 (San Diego Legal Studies, Working Paper No. 10-035, 2010), available at <http://ssrn.com/abstract=1656625> (extending the Ayres and Klemperer model to the context of patent races).

48. See Ayres & Klemperer, *supra* note 46, at 994–1000 (exploring the potential benefits of a patent regime in which remedies for infringement are awarded only a fraction of the time); Sichelman, *supra* note 47 (expanding and analyzing the Ayres and Klemperer model using a variant of quantum game theory).

49. See Ayres & Klemperer, *supra* note 46, at 994–1000 (demonstrating “how uncertainty and delay can produce higher welfare than an ‘idealized’ patent regime”); Sichelman, *supra* note 47, at 14–20 (same).

50. See Burk & Lemley, *supra* note 26, at 1635 (“[R]ewriting the patent law for each industry would involve substantial administrative costs and uncertainties.”); *infra* section II(C)(1).

51. See *infra* subpart II(C)(1).

remedies to change the effect of a substantive rule may be a superior alternative to modifying the substantive rule directly via legislation or regulation.⁵²

In broad terms, I argue that the blanket assumption of courts and scholars that traditional private law remedies are optimal for patent law is flawed. This Article proceeds to do so as follows. Part I describes the aims of patent law, showing that there is nearly universal agreement that the patent system's primary goal is to promote innovation, rather than to vindicate individual, private rights. In so doing, I draw upon scholarly work criticizing patent doctrine for overincorporating traditional contract and private property law concepts, but argue that this line of scholarship is incomplete for failing to critique the incorporation of traditional tort law principles into patent law.⁵³

Part II explains in more detail why existing models of patent law remedies are deficient. In particular, contrary to the prevailing wisdom, I show that injunctions and full compensatory damages may be suboptimal not only for nonpracticing entities but also for practicing entities, in at least four situations: (1) the patent covers a minor component of a complex product and switching costs are high; (2) the costs and risks of innovation are relatively low; (3) large duplicated R&D costs are generated by inefficient rent-seeking; or (4) bargaining costs are far greater than the value of the innovation at issue. Conversely, I argue that awarding more than compensatory damages, even in the absence of willful infringement, may be optimal when infringement is difficult to detect or when the private returns from innovation fall far below the level needed to generate optimal social returns.

Part III addresses possible limitations and criticisms of my proposed model. First, I consider arguments that allowing judges to tailor remedies so as to effectively modify the substantive law threatens traditional separation of powers and related democratic concerns. I conclude that while Congress would need to provide the courts additional discretion to fully eliminate the make-whole approach, the courts have long retained a significant level of equitable discretion to fashion remedies that does not threaten democratic concerns. Second, I address potential concerns that a system unmoored from make-whole remedies would raise innovation costs by increasing overall uncertainty. Relatedly, I examine the concern that judges are insufficiently competent and knowledgeable to impose forward-looking damages in place of injunctions. Partially agreeing with this line of

52. See Lederman & Sichelman, *supra* note 42, at 1685 (proposing a regime of "measured enforcement" wherein the government intentionally engages in imperfect enforcement so as to reduce deadweight losses from taxation).

53. See, e.g., Lemley, *supra* note 25, at 1032 (arguing that intellectual property should not be viewed through the lens of real property law, but overlooking similar problems in classifying intellectual property infringement as a species of tort).

critique, I advocate a near-term regime whereby traditional remedies are kept intact for most situations but are jettisoned in those cases that clearly lead to inefficient results. However, I propose a long-term regime in which remedies in all cases are fashioned with an eye towards optimizing innovation incentives.

Part IV discusses this long-term approach, contending that the remedies framework of the current Patent Act is fundamentally flawed. More generally, I conclude by arguing that the patent system should aim to develop sufficient competence and knowledge in its adjudicators in order to fully discard the notion that patentees should be returned to the *status quo ante*. Instead, patent law remedies should be tailored in each case in order to promote the types and levels of innovation that most benefit society. In this regard, drawing on the work of Shubha Ghosh,⁵⁴ I briefly propose a conceptual model that properly views patent law as a public regulatory mechanism, rather than a system of private law rights, duties, and remedies.

I. The Aim of Patent Law Is Optimally Promoting Innovation, Not Protecting Private Harms

In the United States, the overriding goal of patent law is to promote technological innovation.⁵⁵ In this regard, the Supreme Court has roundly rejected the natural rights notion that patents are a species of personal right designed to protect an inventor's interest in liberty, security, or personhood. As the Court explained at length in 1917 in *Motion Picture Patents*⁵⁶:

Since *Pennock v. Dialogue* was decided in 1829 this court has consistently held that the primary purpose of our patent laws is not the creation of private fortunes for the owners of patents but is to promote the progress of science and useful arts

...

Thirty years later this court, returning to the subject, in *Kendall v. Winsor*, again pointedly and significantly says:

It is undeniably true, that the limited and temporary monopoly granted to inventors was never designed for their exclusive profit or advantage; the benefit to the public or community at large was . . . doubtless the primary object in granting and securing that monopoly.

54. Shubha Ghosh, *Patents and the Regulatory State: Rethinking the Patent Bargain Metaphor After Eldred*, 19 BERKELEY TECH. L.J. 1315 (2004) (proposing a public, regulatory model of patent law).

55. See U.S. CONST., art. I, § 8, cl. 8 (“To promote the Progress of Science and useful Arts” (emphasis added)); *Bilski v. Kappos*, 130 S. Ct. 3218, 3234 (2010) (Stevens, J., concurring) (“[T]he patent system is intended to protect and promote advances in science and technology” (alteration in original) (internal quotation marks omitted)).

56. *Motion Picture Patents Co. v. Universal Film Mfg. Co.*, 243 U.S. 502 (1917).

This court has never modified this statement of the relative importance of the public and private interests involved in every grant of a patent⁵⁷

Nor has the Court wavered from this position since *Motion Picture Patents* was decided.⁵⁸

Moreover, scholars generally agree—not just doctrinally but also as a policy matter—that the foundation of the patent laws should be utilitarian.⁵⁹ This prevailing view draws upon Thomas Jefferson’s apt philosophy, which—in the Supreme Court’s rendering in *Graham v. John Deere Co.*⁶⁰—posited that “[t]he patent monopoly was not designed to secure to the inventor his natural right in his discoveries. Rather, it was a reward, an inducement, to bring forth new knowledge.”⁶¹ Because the patent system rests on a utilitarian basis, as the Court explained in *Kendall v. Winsor*,⁶² any *private* benefit that an inventor receives from a patent is merely a means to an end of providing a benefit “to the public or community at large.”⁶³ As such, any form of private law right afforded to the patentee is

57. *Id.* at 510–11 (citations omitted) (internal quotation marks omitted). Of course, some aspects of patent law promote individual interests at the expense of utilitarian aims. See generally Tom G. Palmer, *Are Patents and Copyrights Morally Justified? The Philosophy of Property Rights and Ideal Objects*, 13 HARV. J.L. & PUB. POL’Y 817 (1990) (proposing a Lockean labor theory approach to patent law). Yet, the primary goal—and the dominant explanation—of U.S. patent law is utilitarian in approach. See Menell, *supra* note 5, at 130 (“The United States Constitution expressly conditions the grant of power to Congress to create patent and copyright laws upon a utilitarian foundation.”); *infra* note 59 and accompanying text.

58. See *Quanta Computer, Inc. v. LG Elecs., Inc.*, 553 U.S. 617, 626–28 (2008) (applying *Motion Picture* to exhaustion of patent rights following an unconditional sale of a patented product); *Kewanee Oil Co. v. Bicron Corp.*, 416 U.S. 470, 480 (1974) (observing that patent laws promote “progress by offering a right of exclusion for a limited period as an incentive to inventors to risk the often enormous costs in terms of time, research, and development” and thus engender a “positive effect on society through the introduction of new products and processes of manufacture into the economy, and the emanations by way of increased employment and better lives for our citizens”); *Mercoird Corp. v. Mid-Continent Inv. Co.*, 320 U.S. 661, 665 (1944) (“It is the public interest which is dominant in the patent system.”); *cf.* *Mazer v. Stein*, 347 U.S. 201, 219 (1954) (“The copyright law, like the patent statutes, makes reward to the owner a secondary consideration.” (internal quotation marks omitted)).

59. See Menell, *supra* note 5, at 130 (“[T]he principal philosophical theory applied to the protection of utilitarian works—that is, technological inventions—has been utilitarianism.”); see also Alan Devlin & Neel Sukhatme, *Self-Realizing Inventions and the Utilitarian Foundations of Patent Law*, 51 WM. & MARY L. REV. 897, 901 (2009) (“Almost all commentators and judges agree that utilitarian considerations enjoy hegemonic status in patent jurisprudence, such that the purpose of the patent system is to induce the creation and commercialization of technology that otherwise could be easily appropriated.”).

60. 383 U.S. 1 (1966).

61. *Id.* at 9; see also Devlin & Sukhatme, *supra* note 59 (“As a patent monopoly carries with it potentially large social welfare costs, it can be condoned only when necessary to incentivize the creation and dissemination of disproportionately valuable information.”).

62. 62 U.S. (21 How.) 322 (1858).

63. *Id.* at 327–28.

purely incidental to the aim of patent law in promoting innovation.⁶⁴ By implication, the vindication of patent law rights via litigation is not meant to remedy a personal wrong inflicted on the inventor.⁶⁵

Thus, patent law rights stand in stark contrast to traditional private law rights, such as those that sound in real property, contract, or tort.⁶⁶ At least historically, the vindication of these kinds of private law rights protected *individual* interests, and these rights grew out of a common law tradition not so much concerned with utilitarian values,⁶⁷ but instead personal autonomy and liberty.⁶⁸ So when a trespasser invaded an owner's land, the owner—to the extent he had Blackstonian property rights that afforded him

64. See La Belle, *supra* note 19, at 50 (“Simply put, patents are a means to an end. And it is that end—the promotion of innovation for the public good—that is paramount.” (footnotes omitted)); cf. Wendy J. Gordon, *Of Harms and Benefits: Torts, Restitution, and Intellectual Property*, 21 J. LEGAL STUD. 449, 450 (1992) (“Uncompensated use of an inexhaustible good is worth discouraging only as a means to an end: obtaining adequate incentives for the good’s initial production and maintenance.”).

65. Of course, one might argue that the patent laws should as a normative matter protect private interests. See ROBERT P. MERGES, JUSTIFYING INTELLECTUAL PROPERTY 14–19 (2011) (espousing a theory of intellectual property based on the labor desert theory of Locke and deontological approach of Kant); Adam D. Moore, *A Lockean Theory of Intellectual Property*, 21 HAMLINE L. REV. 65, 66 (1997) (arguing “that incentives based rule-utilitarian arguments fail to justify anything remotely close to modern Anglo-American copyright, patent, and trade secret institutions”); see also Justin Hughes, *The Philosophy of Intellectual Property*, 77 GEO. L.J. 287, 296–330 (1988) (describing intellectual property according to a Lockean labor theory). And, of course, these approaches would generally counsel in favor of retaining make-whole remedies. However, consistent with the views of the vast majority of courts and scholars, including my own views, I only consider utilitarian approaches herein. See ROBERT P. MERGES, PETER S. MENELL & MARK A. LEMLEY, INTELLECTUAL PROPERTY IN THE NEW TECHNOLOGICAL AGE 11 (5th ed. 2010) [hereinafter MERGES ET AL., INTELLECTUAL PROPERTY] (remarking that utilitarianism is “the dominant paradigm for analyzing and justifying the various forms of intellectual property protection”); John M. Golden, *Principles for Patent Remedies*, 88 TEXAS L. REV. 505, 509 (2010) (“For purposes of simplicity and, in many quarters, plausibility, I generally assume a utilitarian goal that is standard in modern accounts: the patent system should act to promote the development, disclosure, and use of new technologies, ideally in a way that maximizes social welfare.”); Sichelman, *supra* note 35, at 379 (explaining that the foundation of U.S. patent law is utilitarian in nature).

66. Even Merges admits as much. See ROBERT P. MERGES ET AL., INTELLECTUAL PROPERTY, *supra* note 65, at 133–34 (noting that patent rights are modeled on utilitarian, not deontological, principles).

67. See Francis H. Bohlen, *Voluntary Assumption of Risk*, 20 HARV. L. REV. 14, 14 (1907) (noting “the individualistic tendency of the common law . . . proceeding from the people and asserting their liberties”); cf. Roscoe Pound, *Do We Need a Philosophy of Law?*, 5 COLUM. L. REV. 339, 346 (1905) (“Men have changed their views as to the relative importance of the individual and of society; but the common law has not. . . . We no longer hold that society exists entirely for the sake of the individual. . . . The common law, however, is concerned . . . with individual rights.”).

68. See 1 WILLIAM BLACKSTONE, COMMENTARIES *139 (“So great . . . is the regard of the law for private property, that it will not authorise the least violation of it; . . . not even for the general good of the whole community.”); Thomas Merrill, *Property and the Right to Exclude*, 77 NEB. L. REV. 730, 752 (1998) (“[T]he hallmark of property . . . is an individual entitlement grounded in state law, which cannot be removed except for cause.” (internal quotation marks omitted)).

“sole and despotic dominion” over his land⁶⁹—was entitled to exclude the trespasser under all circumstances in order to vindicate his *individual* ownership interests.⁷⁰ Any sort of exception to this rule—for instance, in the event the trespasser crossed onto the owner’s land in order to avoid death⁷¹—was typically a narrow gloss to the individualistic nature of the underlying right.⁷²

More recently, private law scholars have embraced more public-oriented, utilitarian approaches to contract, property, and tort law, particularly in the field of law and economics.⁷³ Yet, as I shall describe, patent law has oddly clung to the more Blackstonian notion of private law. Oddly—unlike the recent law-and-economics accounts of private law—from its constitutional origins, patent law in the United States has been decidedly utilitarian.⁷⁴ In this sense, the patentholder is more akin to a

69. 2 WILLIAM BLACKSTONE, COMMENTARIES *1–2 (“There is nothing which so generally strikes the imagination, and engages the affections of mankind, as the right of property; or that sole and despotic dominion which one man claims and exercises over the external things of the world, in total exclusion of the right of any other individual . . .”); *see also* OLIVER WENDELL HOLMES, JR., THE COMMON LAW 222 (Harv. Univ. Press 2009) (1881) (“The owner is allowed to exclude all, and is accountable to no one.”). Jeremy Bentham, a leading founder of utilitarianism, decidedly criticized the Blackstonian approach. Bentham wrote of “the universal inaccuracy and confusion which seemed to my apprehension to pervade the whole” of the work. JEREMY BENTHAM, A FRAGMENT ON GOVERNMENT 4 (J.H. Burns & H.L.A. Hart eds., Cambridge Univ. Press 1988) (1776).

70. *See* Carl J. Circo, *Does Sustainability Require a New Theory of Property Rights?*, 58 U. KAN. L. REV. 91, 105 (2009) (“Blackstone’s famous reference to an owner’s absolute dominion generally comports with the centrality of individual rights in property as a common law tenet.”); Shyamkrishna Balganesh, *Demystifying the Right to Exclude: Of Property, Inviolability, and Automatic Injunctions*, 31 HARV. J.L. & PUB. POL’Y 593, 598 (2008) (“For quite some time, the right to exclude in the context of both tangible and intangible property has come to be associated with an entitlement to exclusionary (injunctive) relief. Thus, interferences with an owner’s interests are thought to entitle the owner to obtain a permanent injunction restraining such interferences.”).

71. *See* Eduardo Moisés Peñalver & Sonia K. Katyal, *Property Outlaws*, 155 U. PA. L. REV. 1095, 1172 (2007) (“The doctrine of necessity permits nonowners to trespass on, and under certain circumstances even to appropriate, the property of others in order to avoid a grave harm.”); *see also* WILLIAM B. STOEBUCK & DALE A. WHITMAN, THE LAW OF PROPERTY 411 (3d ed. 2000) (“[O]thers [may] enter with the rightful possessor’s permission but [also] in a few cases, against his will.”).

72. *See* Peñalver & Katyal, *supra* note 71, at 1173 (“Many courts . . . have interpreted the necessity defense in an artificially narrow way that would restrict it to extremely unusual circumstances . . .”). *But cf.* Adam Mossoff, *The Trespass Fallacy in Patent Law* 14–16 (George Mason Law & Econ. Research Paper No. 12-54, 2012), *available at* <http://ssrn.com/abstract=2126595> (noting that historical trespass and related disputes were sometimes subject to substantial uncertainty).

73. Abraham Bell & Gideon Parchomovsky, *A Theory of Property*, 90 CORNELL L. REV. 531, 542 (2005) (“[M]ost scholars today base their understandings of property on a model where property is justified by utilitarianism and defined by positive law rather than upon natural rights theories.”).

74. *See* Tun-Jen Chiang, *Fixing Patent Boundaries*, 108 MICH. L. REV. 523, 553–54 (2010) (arguing that the proper mechanism for analyzing the problem of claim amending, and determining on whom to place the burden of loss, is found in the utilitarian purpose of patent law);

private attorney general, paid via the enforcement of his right as a reward for benefitting the *public*, than a vindicator of his own *private* rights.⁷⁵ As many scholars have rightly recognized, instead of conferring private rights on inventors, a public prize system—whereby innovators are merely paid by the State for their endeavors—could, at least in theory, function equally well to promote innovation.⁷⁶ It is for economic reasons, rather than the protection of individual interests, that the United States decided that a patent system was generally more effective than a prize system, patronage, or similar means to encourage innovation.⁷⁷

Yet, judges and scholars came to view the means as the ends. Thus, patents are now viewed not merely as a mechanism the State uses to induce innovation but as a “contract between the inventor and the public,” affording private rights and remedies to the inventor–contractee.⁷⁸

Devlin & Sukhatme, *supra* note 59, at 897 (“Unlike other forms of intellectual property, patents are universally justified on utilitarian grounds alone.”); John F. Duffy, *Inventing Invention: A Case Study of Legal Innovation*, 86 TEXAS L. REV. 1, 5 (2007) (“[U]nlike many other areas, such as tort and criminal law, the patent system has long been based on utilitarian considerations rather than considerations of fairness or justice.”); Adam Mossoff, *Rethinking the Development of Patents: An Intellectual History, 1550-1800*, 52 HASTINGS L.J. 1255, 1315 (2001) (“One common interpretation of eighteenth-century patent doctrine is that patents evolved according to a utilitarian framework with the inventor being granted a reward in exchange for the benefit that accrued to the public.”); David S. Olson, *Taking the Utilitarian Basis for Patent Law Seriously: The Case for Restricting Patentable Subject Matter*, 82 TEMP. L. REV. 181, 182–83 (2009) (“There is widespread agreement that the reason we have a patent system is utilitarian—to solve a market failure problem. . . . Accordingly, a properly crafted patent law should provide enough property rights to incentivize the socially desirable (efficient) level of innovation, and no more.”).

75. See Olatunde C.A. Johnson, *Beyond the Private Attorney General: Equality Directives in American Law*, 87 N.Y.U. L. REV. 1339, 1347 (2012) (explaining that the private attorney general model enables “Congress [to] vindicate important public policy goals by empowering private individuals to bring suit”); see also Daniel Guttman, *Public Purpose and Private Service: The Twentieth Century Culture of Contracting Out and the Evolving Law of Diffused Sovereignty*, 52 ADMIN. L. REV. 859, 917–18 (2000) (“While the tradition of third party government traced here dates to the early twentieth century, there is a much older tradition of the use of private attorneys general or bounty hunters to vindicate public rights, acting in the name of the king or the sovereign.”).

76. See, e.g., Michael Abramowicz, *Perfecting Patent Prizes*, 56 VAND. L. REV. 115, 236 (2003) (“With a carefully designed prize system, even if an individual decisionmaker makes an error or is influenced by political considerations in calculating a prize, these flaws will not affect the decisions that matter—those made in anticipation of the eventual governmental awards.”); Joseph E. Stiglitz, *Economic Foundations of Intellectual Property Rights*, 57 DUKE L.J. 1693, 1721 (2008) (“[I]n areas where there are well-defined needs . . . both the prize system and the patent system can provide comparable incentives to undertake research.”).

77. See *infra* subpart II(A).

78. *Davis Airfoils, Inc. v. United States*, 124 F. Supp. 350, 352 (Ct. Cl. 1954) (“A patent is a contract between the inventor and the public, the terms of which are formulated by the United States Patent Office. The inventor in such a contract gives as a consideration to the public a new and useful art, machine or composition of matter, and, in return, the public gives as a consideration to the inventor a monopoly expressed by the claims of the patent of a period limited by statute”); see also Ghosh, *supra* note 54, at 1316 (“Patents are commonly understood as a hypothetical contract between the inventor and the government resulting in a quid pro quo of innovation for exclusivity.”).

Similarly, instead of treating patents as property-like rights used by inventors to collect payment from society as inducement for their innovative efforts, patents are viewed at root as property rights.⁷⁹ And, last, patent infringement is not just a mode of cataloguing which third parties should provide recompense to the inventor so as to guarantee the appropriate level of R&D and commercialization incentives but rather a species of common law tort.⁸⁰ Judge Frank Easterbrook encapsulates this sentiment when he writes, “Patents give a right to exclude, just as the law of trespass does with real property.”⁸¹

This reification of a patent right as fundamentally a private law right, with all its attendant remedies, is not only conceptually flawed but leads to pernicious outcomes. Several scholars have recognized as much for the contract and real property law models of patents. Shubha Ghosh has challenged the soundness of treating patents as contracts with the State.⁸² In particular, he argues that the “patent law consists of more than a bargain between the inventor and the state; it includes other parties, such as consumers of the invention, follow-on inventors, students of the knowledge created by the innovation, and disseminators of the invention through market and non-market processes.”⁸³ Mark Lemley has convincingly dismissed patent law’s reliance on real property, particularly the notion that patentholders should be able to exclude infringers at-will through injunctive relief.⁸⁴

79. See *Festo Corp. v. Shoketsu Kinzoku Kogyo Kabushiki Co.*, 535 U.S. 722, 730 (2002) (“The patent laws ‘promote the Progress of Science and useful Arts’ by rewarding innovation with a temporary monopoly. The monopoly is a property right . . .” (quoting U.S. CONST., art. I, § 8, cl. 8)); *Zoltek Corp. v. United States*, 58 Fed. Cl. 688, 696 (Fed. Cl. 2003) (“The Federal Circuit, its predecessor court, the Court of Claims, and the U.S. Supreme Court have repeatedly recognized that patent rights are property rights.”); John F. Duffy, Comment, *Intellectual Property Isolationism and the Average Cost Thesis*, 83 TEXAS L. REV. 1077, 1078 (2005) (“I, along with others, believe that intellectual property should be treated as a species of property.”); Adam Mossoff, *Exclusion and Exclusive Use in Patent Law*, 22 HARV. J.L. & TECH. 321, 322 (2009) (“The status of patents is undisputed: patents are property.”); Adam Mossoff, *Patents as Constitutional Private Property: The Historical Protection of Patents Under the Takings Clause*, 87 B.U. L. REV. 689, 690 (2007) (“Patents are property.”); Mossoff, *supra* note 74, at 1255–58 (tracing the origins of the incorporation of private law concepts, such as property and contract, into patent law).

80. *Mars, Inc. v. Coin Acceptors, Inc.*, 527 F.3d 1359, 1365 (Fed. Cir. 2008) (“Patent infringement is a tort.”).

81. Frank H. Easterbrook, *Intellectual Property Is Still Property*, 13 HARV. J.L. & PUB. POL’Y 108, 109 (1990); see also Orin S. Kerr, *Rethinking Patent Law in the Administrative State*, 42 WM. & MARY L. REV. 127, 132–53 (2000) (setting forth a “private law theory” of patent law).

82. See Ghosh, *supra* note 54, at 1339.

83. *Id.*; see also La Belle, *supra* note 19, at 43 (“[P]atent validity challenges are complaints about government conduct that implicate important public interests and potentially affect many parties not before the court. Validity disputes, therefore, fit within [the] paradigm of public law litigation.”).

84. See Lemley, *supra* note 25, at 1046–65 (concluding that granting intellectual property rights imposes complex economic costs, which can only be justified to the extent necessary to

Despite critiques such as Ghosh's and Lemley's, scholars have not recognized—at least in any systematic fashion—that classifying infringement as a common law tort is just as problematic as viewing patents as private contracts or a form of real property.⁸⁵ Indeed, courts and scholars have promoted infringement as a form of tort.⁸⁶ An illustrative example is the Supreme Court's view of patents in 1915 from *Dowagiac Manufacturing Co. v. Minnesota Moline Plow Co.*,⁸⁷ in which it remarked that “[a]s the exclusive right conferred by the patent was property and the

provide incentives to innovate); see also Daniel A. Crane, *Intellectual Liability*, 88 TEXAS L. REV. 253, 253–54 (2009) (arguing intellectual property should not be treated as real property under the law); Mark A. Lemley, *Romantic Authorship and the Rhetoric of Property*, 75 TEXAS L. REV. 873, 895–96, 902 (1997) (book review) (describing the emergence of the property view of intellectual property and arguing against the propertization of intellectual property rights).

85. Lemley has rejected tort law as a model for intellectual property, not because of its compensatory aims but because of a fear that it would lead to restitutionary remedies that focus on the alleged infringer, rather than the IP rights holder. He explains:

In one sense, treating intellectual property as a form of tort law is consistent with the economic lessons of the previous Parts. A focus on harm to the intellectual property owner, rather than on the benefit conferred on the infringer, is consistent with optimal intellectual property policy. . . .

. . . . [But] my fear . . . is that drawing too close an analogy to the tort system will encourage the courts to focus attention on how the defendant was enriched, not on the need for compensating intellectual property owners.

Lemley, *supra* note 25, at 1072. So Lemley in essence embraces the traditional compensatory aim of tort law as it applies to patent remedies, which is consistent with his views in his later work. See *infra* subpart II(B). Indeed, he rejects a public regulatory model of IP. Lemley, *supra* note 25, at 1074 (“Regulation is out of vogue, and those who talk about intellectual property as regulation usually do so in order to denigrate it.”).

86. See, e.g., *In re Cambridge Biotech Corp.*, 186 F.3d 1356, 1371 (Fed. Cir. 1999) (“Patent infringement is properly classified as a tort . . .”); Blair & Cotter, *supra* note 19 (arguing that patent infringement is best characterized as a tort and therefore proximate cause is an appropriate way to determine damages); Roger D. Blair & Thomas F. Cotter, *Strict Liability and Its Alternatives in Patent Law*, 17 BERKELEY TECH. L.J. 799, 800 (2002) (“Patent infringement is often characterized as a strict liability tort . . .”); Kenneth W. Dam, *The Economic Underpinnings of Patent Law*, 23 J. LEGAL STUD. 247, 255 (1994) (“Remedies for infringement of a patent are, with limited exceptions, those appropriate for property.”); Heald, *supra* note 19 (“In establishing what constitutes infringement and what remedies apply, patent law’s secondary function looks like tort law.”); Christopher M. Newman, *Patent Infringement as Nuisance*, 59 CATH. U. L. REV. 61, 67–68, 115–18 (2009) (comparing in economic terms patent infringement to traditional nuisance); Michael L. Rustad, *Torts as Public Wrongs*, 38 PEPP. L. REV. 433, 514 (2011) (“Patent and copyright have long been regarded as creatures of federal statute, but infringement was historically considered to be a tort. For instance, the infringement of a patent was classifiable as a tort—namely, trespass on the case.” (footnotes omitted)); Sandrik, *supra* note 4, at 99 (proposing an “infringement-as-trespass framework” as a basis for understanding patent infringement); cf. Dan L. Burk, *The Trouble with Trespass*, 4 J. SMALL & EMERGING BUS. L. 27, 53 (2000) (arguing that nuisance is an appropriate framework for addressing certain cyberspace disputes); Avihay Dorfman & Assaf Jacob, *Copyright as Tort*, 12 THEORETICAL INQUIRIES L. 59 (2011) (proposing a tort framework for understanding copyright infringement); Mossoff, *supra* note 72, at 11–12 (rejecting the analogy of infringement to pure trespass but proposing a complex analogy based on the violation of various real property interests).

87. 235 U.S. 641 (1915).

infringement was a tortious taking of a part of that property, the normal measure of damages was the value of what was taken.”⁸⁸

As I argue in more detail in Part II, a major problem with viewing patent infringement as a tort is that the private law remedies usually associated with tort law—injunctions and compensatory damages—are not always sensible for optimally encouraging innovation. Rather, I show that in many situations, denying injunctions—even to practicing patentees—and limiting damages to levels below that needed for full compensation may better promote the utilitarian aims of patent law.⁸⁹ Akin to Lemley’s critique of the view that a patent is a form of real property that should allow its owner to internalize the full social value of the invention,⁹⁰ my critique contends that a patent should not necessarily allow its owner to collect the full value of the damages inflicted by an infringer.⁹¹ In yet other situations, I argue that limiting the patentee’s remedy merely to make-whole damages may be insufficient—such as when the overall social value of an invention far exceeds its economic value to private actors in the marketplace.⁹² In these cases, more than make-whole damages—supplied in part from the public fisc—may be a more sensible means of optimizing innovation incentives.⁹³ In order to arrive at these conclusions, I first critique the patent system’s current remedy scheme.

II. Reconfiguring Patent Law Remedies to Promote Innovation

The standard approach to patent law remedies has followed the dominant view that patent law is a form of private law.⁹⁴ Thus, a patentee’s “right to exclude” has historically been treated by courts as akin to a landowner’s right to keep out trespassers, leading judges to grant injunctive relief as a matter of course to prevent ongoing infringement.⁹⁵ Relatedly,

88. *Id.* at 648.

89. See *infra* section II(C)(1).

90. Lemley, *supra* note 25, at 1046 (“The assumption that intellectual property owners should be entitled to capture the full social surplus of their invention runs counter to our economic intuitions in every other segment of the economy. We do not permit producers to capture the full social value of their output. Nor do we permit the owners even of real property to internalize the full positive externalities associated with their property.”).

91. See *infra* section II(C)(1).

92. See *infra* section II(C)(2).

93. See *infra* section II(C)(2).

94. See, e.g., *Carbice Corp. of Am. v. Am. Patents Dev. Corp.*, 283 U.S. 27, 33 (1930) (“Infringement, whether direct or contributory, is essentially a tort, and implies invasion of some right of the patentee.”); *Pall Corp. v. Micron Separations, Inc.*, 66 F.3d 1211, 1221 (Fed. Cir. 1995) (“[P]atent infringement is a continuing tort”); *Rite-Hite Corp. v. Kelley Co.*, 56 F.3d 1538, 1578 (Fed. Cir. 1995) (Newman, J., concurring in part and dissenting in part) (“Patent infringement is a commercial tort, and the remedy should compensate for the actual financial injury that was caused by the tort.”).

95. See, e.g., *Richardson v. Suzuki Motor Co.*, 868 F.2d 1226, 1246–47 (Fed. Cir. 1989) (“Infringement having been established, it is contrary to the laws of property, of which the patent

like an injured party in tort, Congress has legislated, and courts have held, that patentees are entitled to be “made whole” for any past infringement, with damages compensating them for the full extent of the “harm” inflicted upon them by an infringer.⁹⁶ Thus, manufacturing patentees are entitled under current law to a full measure of “lost profits”—and licensing patentees to “reasonable royalties”—in order to compensate them as they would have been paid in a market scenario in which the patent was not infringed.⁹⁷

In the past decade, there has been an assault on the practice of treating a patent as a form of absolute property entitling its holder to injunctive relief.⁹⁸ In large part, this reaction has been driven by the rise of nonpracticing entities (NPEs), particularly the proliferation of lawsuits filed on patents perceived to be “weak,” because they are likely to be held not infringed, invalid, or unenforceable.⁹⁹ Despite the seemingly questionable nature of these patents, critics argue that many of these suits have led to relatively large nuisance-value settlements given the high costs and risks of

law partakes, to deny the patentee’s right to exclude others from use of his property.”); Easterbrook, *supra* note 81 (“Patents give a right to exclude, just as the law of trespass does with real property. Intellectual property is intangible, but the right to exclude is no different in principle from General Motors’ right to exclude Ford from using its assembly line . . .”).

96. See Landers, *supra* note 18 (“[P]atent damages are a make-whole remedy, intended to restore the patentee to the same position as before infringement.”); John W. Schlicher, *Measuring Patent Damages by the Market Value of Inventions—The Grain Processing, Rite-Hite, and Aro Rules*, 82 J. PAT. & TRADEMARK OFF. SOC’Y 503, 503 (2000) (“[P]atent damages should award patent owners the market values or economic benefits of the inventions *that they were unable to capture* due to infringement.” (emphasis added)).

97. See *Aro Mfg. Co. v. Convertible Top Replacement Co.*, 377 U.S. 476, 507 (1964) (“[T]hat question [is] primarily: had the Infringer not infringed, what would Patent Holder-Licensee have made?” (second alteration in original)); *Yale Lock Mfg. Co. v. Sargent*, 117 U.S. 536, 552 (1886) (stating that a patentee’s damages are “the difference between his pecuniary condition after the infringement, and what his condition would have been if the infringement had not occurred”). To be certain, there is some circularity in determining a reasonable royalty. See Shapiro, *supra* note 12, at 308 (“[C]ircularity arises because reasonable royalties are often based on the royalties actually negotiated in the shadow of litigation, and these negotiated royalties depend . . . upon the magnitude of damages that courts are expected to award if the parties are unable to sign a licensing deal and instead engage in patent litigation.”).

98. See, e.g., *eBay Inc. v. MercExchange, L.L.C.*, 547 U.S. 388, 392–93 (2006) (rejecting “invitations to replace traditional equitable considerations with a rule that an injunction automatically follows a determination that a copyright has been infringed”); Landers, *supra* note 19 (contending that patent remedies should differ as between practicing and nonpracticing entities because each incurs different kinds of harm); Lemley & Shapiro, *supra* note 19 (arguing that courts should consider withholding injunctive relief for nonpracticing entities when only one of many different components that make up an infringing product is the subject of the patent-in-suit).

99. *eBay*, 547 U.S. at 397 (Kennedy, J., concurring) (describing the “potential vagueness and suspect validity” of some of “the burgeoning number of patents over business methods”); see also *Patent Quality Improvement: Hearing Before the Subcomm. on Courts, the Internet, and Intellectual Prop. of the H. Comm. on the Judiciary*, 108th Cong. 21 (2003) (testimony of David Simon, Chief Patent Counsel for Intel Corporation) (categorizing NPEs as “patent system bottom feeders” who buy “improvidently-granted patents from distressed companies for the sole purpose of suing legitimate businesses”).

litigation for defendants.¹⁰⁰ Commentators have reasoned that if NPEs only licensed their patents in the marketplace—by definition, they do not manufacture and sell products—then returning them to the *status quo ante* via a patent remedy need only compensate them for lost royalties.¹⁰¹ Thus, the standard remedy of providing injunctive relief to prevent ongoing infringement to NPEs is, on this view, unnecessary to fulfill the private law vision of patent remedies as “making the patentee whole.”¹⁰² On this ground, Justice Kennedy concurred in *eBay* to argue that NPEs typically should not be entitled to injunctive relief.¹⁰³

This widely followed concurrence has led to a substantial reduction in the grant of injunctive relief to NPEs.¹⁰⁴ Yet, as I argued earlier, Justice Kennedy relied on the faulty assumption that make-whole private law remedies optimally promote innovative activity.¹⁰⁵ As I explain in more detail in this Part, in many situations, make-whole remedies may impede efficient levels of innovation.¹⁰⁶ For instance, when a patent merely covers a component of a larger product and the infringer’s switching costs are high, I contend that not only NPEs *but also manufacturing entities* should regularly be denied injunctive relief.¹⁰⁷ Thus, in order to fully excise the

100. See James Bessen & Michael J. Meurer, Essay, *The Direct Costs from NPE Disputes*, 99 CORNELL L. REV. 387, 388–91 (2014) (contending that NPEs resulted in approximately \$29 billion in direct costs for accused infringers in 2011); Brian J. Love & James C. Yoon, *Expanding Patent Law’s Customer Suit Exception*, 93 B.U. L. REV. 1605, 1607–09 (2013) (discussing “bottom feeder” NPEs that file “nuisance-value patent suits”); Michael J. Meurer, *Controlling Opportunistic and Anti-Competitive Intellectual Property Litigation*, 44 B.C. L. REV. 509, 512–16 (2003) (arguing that NPEs use the threat of injunction and high litigation costs to extract exorbitant settlement or licensing fees).

101. See *infra* subpart II(B).

102. See Lemley, *supra* note 6, at 661 (“[W]hat it takes to make the patentee whole is very different if the patentee’s only interest is in licensing the patent than if the patentee’s only interest is in excluding competition and maintaining a monopoly price.” (internal quotation marks omitted)); FTC, *supra* note 15, at 38–40 (noting that NPEs do not produce and sell products and thus are not vulnerable to countersuit, presenting a potential “hold up” threat to the accused infringer).

103. See *eBay*, 547 U.S. at 396 (Kennedy, J., concurring) (“[A]n industry has developed in which firms use patents not as a basis for producing and selling goods but, instead, primarily for obtaining licensing fees For these firms, an injunction, and the potentially serious sanctions arising from its violation, can be employed as a bargaining tool to charge exorbitant fees to companies that seek to buy licenses to practice the patent.”).

104. See *id.* (noting that prior cases established a pattern of granting an injunctive remedy “almost as a matter of course”); FTC, *supra* note 13 (finding that since *eBay*, NPEs have been granted injunctive relief in approximately 45% of the cases in which remedies were awarded and the practicing nature of the party was mentioned).

105. See *supra* notes 11–19 and accompanying text.

106. See *infra* subpart II(B).

107. See *infra* subpart II(B). Recently, in *Apple, Inc. v. Motorola, Inc.*, 869 F. Supp. 2d 901 (N.D. Ill. 2012), Judge Posner denied injunctive relief to Motorola, a practicing entity. *Id.* at 904, 913–15. However, he did so because Motorola had agreed to license the patent-in-issue on fair, reasonable, and nondiscriminatory (FRAND) terms as part of its membership in the standards-setting organization (SSO). See *id.* at 911, 913–14. Thus, denying Motorola injunctive relief

private law specter from patent law remedies, it is not only necessary to remove real property concepts, but also tort concepts, from the doctrinal mix.

In this Part, I begin by recounting the traditional model of patent law remedies. Next, I describe the recent attack on this model for incorporating contract and real property notions. Finally, I argue that in order to complete the attack, it is necessary to counter patent law's reliance on common law tort.

A. *The Standard Model of Patent Law Remedies*

The usual theory explaining why inventors or their employers seek patents assumes that patentees can potentially generate greater-than-average profits on their patented products¹⁰⁸ by preventing others from making, using, and selling those products.¹⁰⁹ According to this theory, society benefits because these supernormal returns compensate for a market defect—namely, that the copying and selling of innovative products by others can often be achieved cheaply and quickly,¹¹⁰ which can suboptimally thwart innovation.¹¹¹ Adopting exclusionary rights via patents

would not thwart its return to the *status quo ante* because Apple had an *ex ante* right to practice Motorola's patent in the marketplace in exchange for a fair and reasonable royalty. *See id.* at 914 ("By committing to license its patents on FRAND terms, Motorola committed to license the '898 to anyone willing to pay a FRAND royalty and thus implicitly acknowledged that a royalty is adequate compensation for a license to use that patent."); *cf.* Crane, *supra* note 84, at 289 ("For example, the rule might be that participation in an SSO automatically leads to the waiver of the right to exclude."). My proposal goes well beyond FRAND situations because it advocates denying injunctions even when the infringer has no right to practice the patent whatsoever. *See infra* subpart II(B).

108. More specifically, patents can also cover methods, processes, and services. *See* 35 U.S.C. § 101 (2006). The reference to "products" in the text is illustrative.

109. *See id.* § 271(a) (2006) (preventing the making, using, and selling of patented products and methods); WILLIAM M. LANDES & RICHARD A. POSNER, *THE ECONOMIC STRUCTURE OF INTELLECTUAL PROPERTY LAW* 294–300 (2003) ("The standard rationale of patent law is that it is an efficient method of enabling the benefits of research and development to be internalized, thus promoting innovation and technological progress."); F.M. SCHERER & DAVID ROSS, *INDUSTRIAL MARKET STRUCTURE AND ECONOMIC PERFORMANCE* 621–24 (3d ed. 1990) (explaining that patent protection prevents competition and allows the inventors to recoup investment).

110. *See* Kenneth J. Arrow, *Economic Welfare and the Allocation of Resources for Invention*, in *THE RATE AND DIRECTION OF INVENTIVE ACTIVITY: ECONOMIC AND SOCIAL FACTORS* 609, 615 (1962) ("In the absence of special legal protection, the owner cannot, however, simply sell information on the open market. Any one purchaser can destroy the monopoly, since he can reproduce the information at little or no cost."); Douglas Gary Lichtman, *The Economics of Innovation: Protecting Unpatentable Goods*, 81 MINN. L. REV. 693, 701–02 (1997) (describing the traditional IP assumption that in the absence of intellectual property protection, new inventions could be easily copied).

111. *See, e.g.,* Rebecca S. Eisenberg, *Patents and the Progress of Science: Exclusive Rights and Experimental Use*, 56 U. CHI. L. REV. 1017, 1024–25 (1989) ("The incentive to invent theory holds that too few inventions will be made in the absence of patent protection because inventions once made are easily appropriated by competitors of the original inventor who have not shared in the costs of invention."); Edmund W. Kitch, *The Nature and Function of the Patent System*, 20

and attendant liability for infringement makes copying costly.¹¹² Thus, the exclusionary rights afforded by patents promote a more optimal level of innovation by providing greater incentives to innovators to invent, market, and sell innovative products,¹¹³ as well as to disclose the knowledge underlying those innovations in the form of published patent documents.¹¹⁴ Importantly, “[t]his traditional conception requires exclusivity; the value of a patent is accordingly commensurate with the value of the market or market niche it controls.”¹¹⁵

Taking the traditional account of patent law on its face, the standard view holds that patent damages should return the patentee to the hypothetical state of affairs that would have obtained but for the infringement.¹¹⁶ For infringement prior to suit, of course, the only possible remedy is some form of monetary payment sufficient to “compensate patentees for their losses.”¹¹⁷ Presuit damages divide into two camps: lost profits and reasonable royalties.¹¹⁸ Lost profits are available to

J.L. & ECON. 265, 276–77 (1977) (arguing that the patent system promotes innovation by allowing inventors to invest in R&D without the fear of their discoveries being copied without their permission); Lichtman, *supra* note 110 (noting that if others could easily copy inventions, “few would want to be innovators, preferring instead to wait and free-ride on someone else’s good idea”).

112. See Lichtman, *supra* note 110, at 700–02 (explaining that under the traditional view, exclusionary rights increase the cost of copying by eliminating the ability for copiers to avoid the development costs incurred through innovation by the original creator).

113. See SCHERER & ROSS, *supra* note 109, at 622–23 (using economic models to illustrate how a lack of patent protection leads to innovators internalizing less of the social benefit from innovation); Kitch, *supra* note 111, at 265–67 (espousing a view that patents provide post-patenting incentives to market and distribute inventions).

114. See Martin J. Adelman, *Property Rights Theory and Patent-Antitrust: The Role of Compulsory Licensing*, 52 N.Y.U. L. REV. 977, 982 (1977) (observing that secrecy deprives the public of the full benefit of the invention and may also lead to duplicative R&D efforts); Jason Rantanen, *Peripheral Disclosure*, 74 U. PITT. L. REV. 1, 4 (2012) (“Given the critical role information plays in invention, it is unsurprising that the patent system has long been justified on the ground that it encourages the disclosure . . .”).

115. Lemley, *supra* note 6, at 657; see also David W. Opderbeck, *Patent Damages Reform and the Shape of Patent Law*, 89 B.U. L. REV. 127, 185 (2009) (“The degree of market power possessed by the owner of a patent on a method or process is a measure of the potential rents the patentee can control by virtue of the patent.”).

116. Lemley, *supra* note 6 (“Patent damages are supposed to compensate patent owners for their losses, putting them back in the world they would have inhabited but for infringement.”); see also BLAIR & COTTER, *supra* note 9, at 113–14 (“Irrespective of the fact that the infringer took measures to avoid patent infringement, if she infringed a valid patent, the infringer will be liable for the full economic injuries that the infringement caused. In principle, the patentee will be fully compensated for any injury due to the infringement.”); Blair & Cotter, *supra* note 19, at 44–70 (advocating a “but for” standard of patent remedies).

117. See Lemley, *supra* note 6, at 655 (citing *Pall Corp. v. Micron Separations, Inc.*, 66 F.3d 1211, 1223 (Fed. Cir. 1995)); see also 35 U.S.C. § 284 (2006) (“[T]he court shall award the claimant damages adequate to compensate for the infringement.”).

118. See 35 U.S.C. § 284; Lemley, *supra* note 6, at 655 (“Courts interpreting this provision have divided patent damages into two groups—lost profits . . . and reasonable royalties.”). Disgorgement of the infringer’s profits was another available remedy, but Congress eliminated it

manufacturing patentholders who would have made sales—and associated profits—but for the infringement.¹¹⁹ When the patentee does not practice its patent, or lost profits cannot be proved, reasonable royalties are provided.¹²⁰ To prevent *post*-suit infringement, prior to the Supreme Court's decision in *eBay*,¹²¹ the standard approach was to grant an injunction.¹²² So while nonpracticing entities licensed their inventions in the marketplace, they did so under the threat of an injunction in the event of infringement.¹²³

B. Removing Real Property: A Partially Revised Model of Patent Law Remedies

With the rise in suits from nonpracticing entities (NPEs) in the early 2000s, the standard model of patent remedies was impugned by several leading scholars as too often providing “windfalls” to patentees that typically performed no R&D, and certainly did nothing to commercialize their patents.¹²⁴ In a widely cited article, *Property, Intellectual Property,*

in 1946 because of the perceived difficulty and high costs of determining these profits. *See* 35 U.S.C. § 70 (1940), *amended by* Act of Aug. 1, 1946, ch. 726, 60 Stat. 778 (codified at 35 U.S.C. § 70 (1946)); H.R. Rep. No. 79-1587, at 1–2 (1946). Interestingly, disgorging the infringer's profits would not return the patentee to the *status quo ante* because the patentee's profits in the absence of infringement could be substantially less than the infringer's profits.

119. *See* *Panduit Corp. v. Stahl Bros. Fibre Works, Inc.*, 575 F.2d 1152, 1156 (6th Cir. 1978) (Markey, C.J.) (“[Damages] have been said to constitute ‘the difference between his pecuniary condition after the infringement, and what his condition would have been if the infringement had not occurred.’”) (quoting *Aro Mfg. Co. v. Convertible Top Replacement Co.*, 377 U.S. 476, 507 (1964) (quoting *Yale Lock Mfg. Co. v. Sargent*, 117 U.S. 536, 552 (1886))).

120. *See id.* at 1157 (“When actual damages, e.g., lost profits, cannot be proved, the patent owner is entitled to a reasonable royalty.”); *see also* Lemley, *supra* note 6, at 671 (“Patentees whose harm is based on a loss of market exclusivity—those who reasonably could have expected to make additional sales, or sales at a higher price, absent infringement—should be entitled to lost profits damages. Patentees whose harm is lost licensing revenue, but who could not plausibly claim to have lost sales as a result of the infringement, should be entitled to reasonable royalties . . .”).

121. *See eBay Inc. v. MercExchange, L.L.C.*, 547 U.S. 388, 394 (2006) (holding that injunctions are not mandatory but should instead be awarded on the basis of a set of equitable factors).

122. *See id.* at 394–95 (Roberts, C.J., concurring) (noting the historical practice of granting injunctions in most patent cases); Lemley & Shapiro, *supra* note 19; *cf.* *Smith Int'l., Inc. v. Hughes Tool Co.*, 718 F.2d 1573, 1577–78 (Fed. Cir. 1983) (stating that without the ability to issue injunctions, the “right to exclude granted by the patent would be diminished, and the express purpose of the Constitution and Congress, to promote the progress of the useful arts, would be seriously undermined”).

123. *See supra* notes 100–03 and accompanying text.

124. *See, e.g.,* Amy L. Landers, *Let the Games Begin: Incentives to Innovation in the New Economy of Intellectual Property Law*, 46 SANTA CLARA L. REV. 307, 308–09 (2006) (“At their worst, patent trolls threaten companies with baseless lawsuits that seek product shutdowns and large monetary judgments.”); Lemley & Shapiro, *supra* note 19, at 2009 (“These companies are paying holdup money to avoid the threat of infringement. That is not a legitimate part of the value of a patent; it is a windfall to the patent owner that comes at the expense not of unscrupulous copyists but of legitimate companies doing their own R&D.”).

and Free Riding, Mark Lemley derides the real property foundation implicit in patent law's exclusion principle.¹²⁵ Specifically, Lemley begins his critique by returning to the fundamental goal of patent law: "to give as little protection as possible consistent with encouraging innovation."¹²⁶ In this vein, he argues that "[t]he absolute protection or full-value view draws significant intellectual support from the idea that intellectual property is simply a species of real property rather than a unique form of legal protection designed to deal with public goods problems."¹²⁷ In particular, Lemley rejects the view that a patent owner should be able to internalize the full social value of his invention because an inventor need not capture this entire value in order to be optimally incentivized to invent.¹²⁸ Rather, sufficient incentive will be present as long as the inventor earns a suitable return on its investment in R&D.¹²⁹

Yet, oddly, when it comes to specific remedies, Lemley essentially limits the implications of his critique to those entities that non-exclusively license their patents, such as nonpracticing entities.¹³⁰ For instance, he concludes that "if the baseline assumption of the law is that the intellectual property owner is entitled to capture the full social value of the invention, it is that baseline that will drive any *licensing* negotiations."¹³¹ He advocates routinely denying injunctions to nonpracticing entities in situations involving small components of complex products (so-called "component patents") and high switching costs, but he does not do so for practicing entities, instead suggesting that injunctive relief is generally appropriate.¹³² However, as I argue below, Lemley's same baseline assumption and related critique hold true for the value the manufacturing patentee is entitled to capture because third parties will gauge the financial cost of infringing based upon the expected damages they will pay in court and the related costs of complying with an injunction.¹³³

125. Lemley, *supra* note 25, at 1031–32.

126. *Id.* at 1031.

127. *Id.* at 1031–32.

128. *See id.* at 1065 ("Economic theory tells us that we must balance those rights if we are to achieve efficiency, granting intellectual property rights only to the extent necessary to enable creators to cover their average fixed costs. Anything more does harm and no good." (footnotes omitted)).

129. *Id.* at 1046 ("In a market economy, we care only that producers make enough return to cover their costs, including a reasonable profit.").

130. *See id.* at 1045–46.

131. *Id.* at 1046 (emphasis added).

132. *See* Mark A. Lemley, *Patenting Nanotechnology*, 58 STAN. L. REV. 601, 630 (2005) ("The ability of patent owners to . . . threaten injunctions against whole products even if a small component is found to infringe has created a thriving economy of patent 'trolls.' These patent trolls game the system in an effort to capture not only the value of their inventions, but the value of complementary assets and irreversible investments made by defendants as well.").

133. *See infra* section II(C)(1).

In a later and highly influential article by Lemley and Carl Shapiro, *Patent Holdup and Royalty Stacking*, these conceptual limitations are even more apparent. Lemley and Shapiro begin their analysis with a convincing argument that the granting of injunctions to nonpracticing entities on component patents can provide a windfall to the patentee that distorts optimal innovation incentives.¹³⁴ Specifically, Lemley and Shapiro determine a “benchmark royalty,” which represents the optimal level of licensing fees that a nonpracticing licensor would earn in the “ideal patent system.”¹³⁵ Although there is no need here to recount all of the mathematical details of how the benchmark royalty is precisely calculated to illustrate my point, importantly, the royalty is a fraction of the economic value of the invention to the licensee (though one might suitably recharacterize it as based on the value of the invention to society).¹³⁶

Unlike Lemley’s earlier work, Lemley and Shapiro do not posit that the benchmark royalty should be the licensee’s pro rata portion of the *minimum* amount necessary to induce invention on the part of the patentee.¹³⁷ However, because the benchmark royalty is generally a fraction of the full economic value of the invention to the licensee, we would expect it to roughly approximate such an amount. Conversely, if the patentee were able to garner substantially *more than* the full economic value of the invention to the licensee, then we would expect that this amount would generally be far greater than that needed to induce invention. Thus, although Lemley and Shapiro do not reject the make-whole damages assumption per se, their theoretical analysis showing that injunctions can sometimes produce gross windfalls for the patentee implicitly supports the thesis that injunctions can thwart optimal innovation incentives.

Suppose, for instance,¹³⁸ that a patent allows a manufacturer to cut its production costs by \$1 per product sold by reducing the costs of

134. See Lemley & Shapiro, *supra* note 19, at 2009 (“These companies are paying holdup money to avoid the threat of infringement. That is not a legitimate part of the value of a patent; it is a windfall to the patent owner that comes at the expense not of unscrupulous copyists but of legitimate companies doing their own R&D.”). The same kind of analysis applies to patent claims covering an entire complex product, the only novel elements of which are components of the product. See Mark A. Lemley, Essay, *Point of Novelty*, 105 NW. U. L. REV. 1253, 1260 (2011) (“[W]e don’t distinguish between the novel elements of the claim and the preexisting ones.”).

135. Lemley & Shapiro, *supra* note 19, at 1999.

136. See *id.* (using the value of the use of the patented invention to the downstream firm (licensee) as the base for determining the benchmark royalty rate).

137. See *id.* at 1999–2000.

138. Here, I provide several numerical examples to support my case. In so doing, I do not intend to show that the traditional analysis is incorrect in *every* case; rather, I do so to show that it is incomplete in a subset of cases that are likely to occur in practice. Moreover, although I present numerical examples, I follow Lemley and Shapiro’s formal algebraic model, which applies in a wide set of circumstances. See *id.* at 1996–2005.

manufacturing a particular component in the product.¹³⁹ In this event, Lemley and Shapiro's benchmark royalty is some fraction of the \$1.¹⁴⁰ For simplicity, if we suppose the licensee is the only potential producer of the product,¹⁴¹ and it sells one million products per year, the total economic value produced by the innovation is \$1 million per year. Also suppose that prior to the inventive technology, the licensee sold the product for \$100 (resulting in \$100 million in annual sales), whereby each sale returned \$9 in cash profits (resulting in \$9 million in annual profits). This profit may derive, for example, from the licensee's patents, know-how, marketing, or other competitive advantages.¹⁴² Thus, by using the patented process, the licensee can increase its profits to \$10 million per year, less any payments to the patentee.

In this example, it is relatively straightforward to see how a mandatory injunction can systematically skew licensing rates or settlement fees. Suppose that the licensee-seller independently invents the cost-saving technology, implements it, and is then sued by the patentee for infringement.¹⁴³ If the patentee can shut down all sales of the infringing

139. Importantly, standards-essential patents—those patents that must be practiced in order to implement an industry-wide standard—often cover small components of complex products. As Suzanne Michel has noted:

Recent, very large damage awards, some of which have been based on one feature of a complex product covered by hundreds of patents and increases in IT industry patent litigation raise questions of whether damages law is sufficiently economically grounded, at least in that subset of cases in which the invention is one component of a complex IT product. Of course, these are the types of cases most likely to involve a patent covering a standard.

Suzanne Michel, *Bargaining for RAND Royalties in the Shadow of Patent Remedies Law*, 77 ANTITRUST L.J. 889, 899 (2011).

140. See Lemley & Shapiro, *supra* note 19, at 1999. Specifically, they find that the benchmark royalty rate, assuming an ideal patent system and parties of equal bargaining power, would be half the gains from using the invention adjusted by the strength of the patent. *Id.* Thus, if there is an 80% chance the patent would be found valid, enforceable, and infringed, and the value of using the patented invention is \$1, the benchmark royalty would be \$0.40 (half of \$1 × 80% probability of infringement). See *id.*

141. For instance, the licensee may own patents covering other components of the product that prevent its manufacture by third parties, including the licensor. See Sichelman, *supra* note 35, at 354 (“[M]any patented products contain components that infringe earlier patents.”); Robert P. Merges, *Intellectual Property Rights and the New Institutional Economics*, 53 VAND. L. REV. 1857, 1859 (2000) (“Complex, multi-component products are the norm in many industries (e.g., autos and consumer electronics), and individual patents often cover only a single component or sub-component.”).

142. See Lemley, *supra* note 6, at 663 (“Even if there are no other relevant patents, the defendant’s know-how, materials, and marketing efforts almost always contribute some value, and usually the most significant part of the value of an infringing product.”); Brian J. Love, Note, *Patentee Overcompensation and the Entire Market Value Rule*, 60 STAN. L. REV. 263, 278 (2007).

143. Recent empirical work indicates that infringers tend not to “copy” from the patentee in the course of infringement. See Christopher A. Cotropia & Mark A. Lemley, *Copying in Patent Law*, 87 N.C. L. REV. 1421, 1424 (2009) (finding that “copying” is proven by accused infringers in less than 2% of all cases and alleged in less than 11%). These figures very likely understate the

product, which it typically can—because the product was made using the infringing production method and the resulting component¹⁴⁴—the seller is faced with two scenarios.¹⁴⁵ First, the seller may be able to profitably redesign the product in a noninfringing manner—for example, by returning to the prior production technique and component. If the redesign is costly—because, for instance, it requires significant retooling of manufacturing processes—say \$2 million amortized over each year of product sales (which includes lost sales of products during the period of redesign), the seller will negotiate a license rate at far more than the optimal one of some fraction of \$1 per product. In particular, the seller reasons that if it redesigns, then it stands to earn \$7 million per year in profit (the original \$9 million per year in profits less the \$2 million per year in retooling costs), whereas if it does nothing, it could lose all profits if it is enjoined from using the patented method. The seller also reasons that if it continues to use the patented production method, it will stand to earn \$10 million per year (the original \$9 million plus the \$1 million in added savings) less any licensing or settlement fees.

If the patent at issue is “ironclad”—meaning it would certainly be found infringed, and not invalid or unenforceable¹⁴⁶—and the parties are of

amount of copying present in the marketplace. First, copying probably more often results from the copying of others’ products, rather than patents, and mere product copying is unlikely to find its way into the kinds of litigation documents Cotropia and Lemley examined. See Golden, *supra* note 65, at 589 n.456 (providing an example of such). Second, based on my personal experience, documentary evidence of direct copying is usually scant, and litigators tend to do an outstanding job in preparing witnesses to avoid disclosing any copying in depositions. Nonetheless, there are surely many cases of independent invention. Moreover, as I argue below, the analysis here can, under the right conditions, apply in the event the accused infringer copied the patented technology. So, although I do not agree that lack of copying inherently implies independent invention, it is likely to do so in a significant number of cases. Moreover, I argue below that injunctions may distort innovation incentives even in the presence of copying. See *infra* note 149.

144. The patentee can typically draft patent claims covering not only the inventive production process but also the resulting component and even the entire product, offering numerous routes to proving infringement. See Landers, *supra* note 124, at 360 (“[P]atentees drafting improvement claims may be encouraged to include additional components in combination claims to sweep additional products within their scope. For example, a patentee who claims an improvement to one component of a computer networking system can mention other components of the system in the language of the claim to later support an argument that the royalty base should include the entire system.”). Nonetheless, even if only the component is covered by the applicable patent claim, an injunction over the component will in many cases effectively result in an injunction over the entire product. See, e.g., Michael A. Carrier, *Innovation for the 21st Century: A Response to Seven Critics*, 61 ALA. L. REV. 597, 619 (2010) (citing *z4 Technologies, Inc. v. Microsoft Corp.* as an example in which “Microsoft infringed [because of] a ‘very small component’ . . . not related to the ‘core functionality’ of Windows in a setting in which injunctive relief would have required Microsoft to release new versions of its Windows software in 600 variations in more than 40 languages”).

145. I have slightly adapted Lemley and Shapiro’s analysis in the following exposition. See Lemley & Shapiro, *supra* note 19, at 2001–05.

146. Lemley & Shapiro, *supra* note 19, at 1999 (implying that an “ironclad” patent is one that is clearly infringed, valid, and enforceable).

equal bargaining power, then the licensee will pay half the expected marginal profits from using the patented production technique relative to the old technique.¹⁴⁷ Here, the marginal difference is \$3 million—namely, \$10 million per year (i.e., profits using the new technique) less \$7 million per year (i.e., profits from retooling to the old technique). So, in an equal bargaining power scenario, the patentee secures \$1.5 million per year, or \$1.50 per product, in royalties. This amount, of course, is more than the entire cost-reducing value of the patented invention. In other words, the threat of an injunction coupled with high switching costs can enable the patentee to extract more than the social value of its invention in rents from the potential user.¹⁴⁸

The situation is even worse if the seller cannot design around the patent. Suppose, for instance, that the seller had just launched its business, only to learn that one of its key product components was patented.¹⁴⁹ Because the patentee can shut down the seller's operations entirely, assuming the parties are of equal bargaining strength, and the patent is ironclad, the marginal difference will be the potential profits of operating (here, \$10 million per year) less the profits of being shut down (of course, nothing—perhaps even negative profits). In this case, the seller must forgo \$10 million in potential profit, and licenses the patent for \$5 million per year, or \$5 per product, in royalties—much more of a distortion from the optimal royalty rate.

In the previous example, only the licensee-seller—but not the patentee—manufactured the relevant product. Yet all of Lemley and

147. See *supra* note 140 and accompanying text.

148. Here, I have assumed that the social value of the patent equals its total cost-saving potential. Of course, the social value of a patent can be greater than its private economic value, in which case we might justify the greater reward to the patentee. See *infra* section II(C)(2). Yet, in the standard make-whole remedies approach, courts assume that the profits or royalties lost by the patentee—i.e., its *private market value* to the patentee—is sufficient to reward the patentee. As a first cut, here I do the same. I relax this assumption below. See *infra* section II(C)(2).

149. For standards-essential patents, it is typically impossible to design around the patent and still practice the standard. See Patrick J. Flinn, *Why FRAND Matters*, INTELL. PROP. & TECH. L.J., Feb. 2013, at 8, 11 (explaining that “when the patent is a standards-essential patent . . . the question is now whether to enter the market at all because license to every standards essential patent is needed”). Thus, even in the absence of a FRAND commitment—and in the presence of direct copying by the accused infringer—for the reasons presented herein, the presumption should generally be to deny injunctions to practicing and nonpracticing entities alike for infringement of standards-essential patents covering minor components of complex products. See Joseph Scott Miller, *Standard Setting, Patents, and Access Lock-In: RAND Licensing and the Theory of the Firm*, 40 IND. L. REV. 351, 378 (2007) (“If the owner of a standard-essential patent can enjoin (or threaten to enjoin) would-be adopters from practicing the standard, the very enterprise of adopting a standard fails to meet its basic purpose.”); cf. Colleen V. Chien et al., *RAND Patents and Exclusion Orders: Submission of 19 Economics and Law Professors to the International Trade Commission 2* (Santa Clara Univ. Sch. of Law Legal Studies Research Paper Series, Paper No. 07-12, 2012), available at <http://ssrn.com/abstract=2102865> (recommending that injunctions generally be denied in International Trade Commission proceedings on all standards-essential patents subject to a FRAND commitment).

Shapiro's arguments and mathematical modeling apply with similar force to practicing entities that compete directly with the infringer. For instance, if a practicing patentee is granted an injunction, just as with a nonpracticing entity, the infringing firm, as Lemley and Shapiro recognize, "cannot sell the infringing product and must withdraw it from the market unless and until the firm can introduce a redesigned version that does not contain the patented feature, or until the patent expires."¹⁵⁰

To illustrate the consequences of this observation, take the same patented invention as the earlier example, but now suppose the patentee also sells the product. Suppose prior to the patented invention, the product sells at \$98, instead of \$100, because the duopolistic market is more competitive than the monopolistic market of the previous example.¹⁵¹ In this event, the infringer earns \$7 per product in profit (as the marginal cost per product prior to the cost-saving invention is \$91). Suppose further the infringer sells half the number of products per year (500,000 instead of the original million). Also suppose that the patentee earns the same profit and sells the same number of products per year as the infringer.¹⁵² Like the original example, this duopoly may arise from other patents owned by the parties or from other barriers to entry.¹⁵³

When the patented cost-saving component comes along, if both parties use it—assuming for simplicity that the market is fully competitive at the margins¹⁵⁴—then the price of the product drops from \$98 to \$97, resulting in \$1 million per year in additional consumer surplus. In the event the patentee is granted injunctive relief, it has two choices. First, it can shut down sales of the competitor's product entirely until the competitor redesigns it.¹⁵⁵ During this period, because the patentee is the only seller in

150. Lemley & Shapiro, *supra* note 19, at 1996.

151. As with the earlier example, this example is generalizable across a host of variables. See generally George Symeonidis, *Comparing Cournot and Bertrand Equilibria in a Differentiated Duopoly with Product R&D*, 21 INT'L J. IND. ORG. 39 (2003) (offering a general model of duopolistic behavior in the presence of R&D).

152. Because the firms sell products that are slightly different in characteristics, see *infra* note 154, under a Bertrand model for differentiated products, firms would typically choose to price at above marginal cost. See DAVID BESANKO & RONALD BRAEUTIGAM, MICROECONOMICS 563 (4th ed. 2011) ("In a Bertrand equilibrium with differentiated products, equilibrium prices generally exceed marginal cost.").

153. See Harold Demsetz, *Barriers to Entry*, 72 AM. ECON. REV. 47, 47–57 (1982) (discussing generally the various kinds of barriers to entry into an economic market); *supra* note 142 and accompanying text.

154. For instance, the \$5 per product profit earned by each party may result from differences in product features attributable to their separate patents, leading to market differentiation, whereas the cost-saving feature merely leads both sellers to cut their product prices by \$1. See Greg Vetter, *Patenting Cryptographic Technology*, 84 CHI.-KENT L. REV. 757, 774–75 (2010) (noting the use of patents for product differentiation).

155. In *Apple Inc. v. Samsung Electronics Co.*, 695 F.3d 1370 (Fed. Cir. 2012), the Federal Circuit denied a manufacturing patentee a preliminary injunction over the accused infringer's product containing a patented component because the patentee failed to show a "causal nexus"

the market, it can sell the product at the monopoly price (here, \$100), leading to increased deadweight losses (here, some fraction of \$3 million per year) that far exceed the social value of the patented invention. When the competitor redesigns its production process, as in the earlier example, it must spend \$2 million per year on an amortized basis, which is more than the value of the patent, and hence a net social loss. Of course, the competitor has a financial incentive to redesign in order to avoid being shut down and earning nothing.

Second, the patentee could decide to license the competitor once it learns of the infringement. In this event, an analysis similar to that recounted in the nonpracticing entity example above applies. Namely, the patentee reasons that the infringer has one of two viable choices: (1) retool and earn \$1.5 million per year in profits (\$3.5 million in gross profits less the \$2 million in amortized retooling costs); or (2) license and earn \$3.5 million per year in profits (or perhaps more if the license allows the parties to effectively fix prices) less any licensing fees. If the patent is ironclad and the parties are of equal bargaining power, then the licensor will demand annual royalties of half the difference between \$3.5 million or more and \$1.5 million (i.e., at least \$1 million), which is equal to or more than the social value of the invention.¹⁵⁶

Thus, just like nonpracticing patentees, following Lemley and Shapiro's otherwise insightful analysis, manufacturing patentees with injunctive threats can—when switching costs are abnormally high—effectively extract or cause systematic deviations from the optimal profits or payment for a given invention, as well as unnecessarily raise consumer deadweight losses if the infringer cannot design around the patent or the design-around requires a long period of time. Lemley and Shapiro, however, provide no significant analysis of the manufacturing entity situation. Instead, they contend without extended discussion that:

[W]e stress that our analysis in this Article is expressly limited to

between the alleged irreparable harm from the accused infringer's sales of the entire product and its infringement of the specific patented component. *Id.* at 1374–75. In other words, because “it may very well be that the accused product would sell almost as well without incorporating the patented feature,” the Federal Circuit found preliminary relief unwarranted. *Id.* In a later ruling in the same case, the Federal Circuit held that the causal nexus rule applies to permanent injunctions. *Apple Inc. v. Samsung Elecs. Co.*, 735 F.3d 1352, 1361 (Fed. Cir. 2013). Nonetheless, in the example in the text above, the rule would not prevent the patentee from securing an injunction because the cost-cutting patented feature here would enable the patentee to garner 100% of the sales of the patented product in the event of no infringement. *See Apple*, 695 F.3d at 1375 (stating that the causal nexus requirement is met when the patentee “show[s] that the infringing feature drives consumer demand for the accused product”). Thus, although the “nexus” requirement adopted in the *Apple v. Samsung* line of cases ameliorates some of the deleterious effects of injunctions in the context of multicomponent products, it does not fully rectify overdeterrence problems.

156. *Cf.* Heald, *supra* note 19, at 1184 (“In fact, if the infringing firm's switching costs are high enough, it may be willing to agree to a license that results in a net loss.”).

situations in which the patent holder's predominant commercial interest in bringing a patent infringement case is to obtain licensing revenues. Our policy recommendations here pertain only to this type of situation, where the patent holder can claim reasonable royalties but not lost profits, and not to settings in which the patent holder suffers significant lost profits as a result of the allegedly infringing activities of the downstream firm and seeks to use the patent to exclude a competitor from the market in order to preserve its profit margins. In cases involving significant lost profits, we favor a presumption that the patent holder will be granted a permanent injunction, perhaps with a stay to allow the infringing firm to redesign its product. The presumptive right to a permanent injunction in these cases is justified in part for reasons of equity and in part because of the grave difficulties associated with calculating and awarding lost profits on an ongoing basis.¹⁵⁷

Lemley and Shapiro's categorical distinction between licensing and manufacturing entities is not economically justified. Specifically, their suggestion of a stay to allow the infringer to redesign the product does not always solve holdup concerns. As I showed earlier, the costs of redesign may exceed the value of the patented component, clearly a net social loss. Additionally, there may be situations for which no redesign is available. In essence, Lemley and Shapiro's resort to "reasons of equity" to justify presumptive injunctions for manufacturing patentees sounds in the same reasoning as that in tort law because it aims to provide an injunction "only when the remedy of compensatory damages will not suffice to restore the status quo ante."¹⁵⁸ But as a matter of pure economics, the simple examples above—which are easily generalizable—show that traditional tort law remedies do not always promote optimal outcomes in the patent law context.

An additional argument commentators have made to distinguish practicing entity suits from NPE suits is that accused infringers can typically assert their own patents back against a practicing patentee in a counterclaim or separate suit, thereby preventing holdup.¹⁵⁹ Although many direct competitors cross-license one another or settle in the face of a potential countersuit, there are numerous disputes in which an accused infringer cannot credibly threaten to shut down a practicing patentee.¹⁶⁰

157. Lemley & Shapiro, *supra* note 19.

158. Fellmeth, *supra* note 4.

159. See FTC, *supra* note 15 ("The potential for hold-up to result in mutually assured destruction means firms actively participating in the industry—patent practicing entities (PPEs)—are unlikely to employ this hold-up strategy against each other.").

160. See, e.g., Ted Sichelman, *The Vonage Trilogy: A Case Study in "Patent Bullying"* 7 (San Diego Legal Studies, Paper No. 11-057, 2013), available at <http://ssrn.com/abstract=>

Specifically, the accused infringer may not own any patents that it can assert against the patentee or may simply own far fewer applicable patents than the patentee plaintiff, causing insufficient threat to the patentee.¹⁶¹

Another simple hypothetical illustrates why the distinction between nonpracticing and practicing entities is problematic—at least if we take at face value the dominant reward theory of patent law, which justifies awarding exclusionary rights on the basis of allowing inventors to recoup their costs with a suitable profit.¹⁶² In one state of the world, inventor A spends \$1 million in R&D, designing, building, and testing a particular invention. The inventor then manufactures the invention and sells one million units at \$100 each, earning \$10 in profit, of which \$1 is attributable to the patented invention (like the earlier example).

Assume, further, that—consistent with the reward theory of patents—manufacturing and commercialization of the product does not incur substantial cost or risk that requires additional patent protection to ensure optimal levels of investment.¹⁶³ In this regard, it is important to recognize that Justice Kennedy decidedly did not adopt a commercialization theory of patents in *eBay*—namely, that NPEs should be punished merely because they fail to practice their patents.¹⁶⁴ Rather, his decision stemmed from his view that NPEs gain “undue leverage” from potential injunctive relief in negotiating with infringers, primarily because NPEs only garner money damages in the marketplace and cannot be countersued in an infringement suit.¹⁶⁵ So while commercialization goals might very well play an important role in patent law—and I have advocated as much elsewhere¹⁶⁶—taking Justice Kennedy and the vast majority of scholars at face value,¹⁶⁷ I assume in this example that commercialization is not an interest of concern.

In the event of infringement—using the same base assumptions as the earlier example—assume further that the patentee’s sales drop in half (from

1856703 (“Vonage’s lack of a patent portfolio (by definition) precluded it from the preferred tactic of ‘defending’ against suit by asserting its patents back against Sprint.”).

161. *See id.*

162. *See* Sichelman, *supra* note 35, at 357–58 (discussing reward theory); Mark A. Lemley, *Ex Ante Versus Ex Post Justifications for Intellectual Property*, 71 U. CHI. L. REV. 129, 129 (2004) (“The traditional economic justification for intellectual property is well known. Ideas are public goods: they can be copied freely and used by anyone who is aware of them without depriving others of their use.”).

163. *See* Lemley, *supra* note 162, at 148–49 (contending that at least one *ex post* justification for intellectual property asserts that commercialization of inventions will tend to proceed efficiently following the invention phase).

164. *See supra* notes 11–16 and accompanying text.

165. *See supra* notes 11–16 and accompanying text.

166. *See* Sichelman, *supra* note 35, at 354 (arguing that commercialization efforts typically entail supernormal costs and risks that independently justify some form of *ex post* IP protection).

167. *See id.* at 344 (“The dominant ‘reward’ theory of patenting, which undergirds much of today’s law, perceives little to no need to protect risky and costly post-invention development and commercialization efforts.”).

one million to 500,000 units) and its profits drop by more than half (from \$10 to \$4 per unit), resulting in only \$2 million in annual profits (500,000 units $\times$ \$4 per unit). Under current law, the inventor is entitled to lost profits, and is able to recoup \$8 million in annual damages for past infringement, returning it to the \$10 million per year in total profits.¹⁶⁸ The net result is that the patentee earns the entire \$1 million per year in additional profits attributable to the patented invention. Yet, in another state of the world, imagine the patentee decides to license its invention. The would-be licensee rejects the offer and decides instead to infringe. The patentee wins at court and now collects a reasonable royalty. Even assuming a high royalty of 25% of the additional profits generated by the patented invention,¹⁶⁹ the patentee only collects \$250,000 in annualized damages.¹⁷⁰

So the practicing inventor spends the \$1 million on R&D and reaps \$1 million per year in benefits, while the nonpracticing inventor spends the same \$1 million and collects \$250,000 in annual royalties. This disparity has no economic basis in the generally accepted view of patent law to provide optimal incentives to invent, rather than commercialize. At least for this hypothetical, the two inventors should earn *exactly the same return* on their efforts. As John Golden properly recognizes in a response to Shapiro and Lemley's article, "A per se rule of discrimination based on a patent holder's business model could act as an undesirable drag on the efficiency and competitiveness of markets for innovation."¹⁷¹ In a reply to Golden, Lemley and Shapiro—instead of engaging in economic analysis—

168. For simplicity, suppose that \$7 million of the \$8 million is awarded for damages relating to other patents. However, as noted below, the patentee is often entitled to damages on the entire market value of a product merely due to a single patented component. Here, the cost-saving component would have put the patentee in a position to make 100% of the sales of the product in the market because it could price below all competitors in the event of no infringement. As such, the patentee could show a nexus between the component and the purchase of the product by consumers, justifying imposition of the entire market value rule. See *infra* note 183 and accompanying text.

169. Under current law, reasonable royalties are typically apportioned to reflect a royalty on the value of the patented component. See *Lucent Techs., Inc. v. Gateway, Inc.*, 580 F.3d 1301, 1339 (Fed. Cir. 2009) (holding that "when the patented invention is a small component of a much larger commercial product," the royalty base may be the sale price of the entire product "so long as the [royalty] multiplier accounts for the proportion of the base represented by the infringing component or feature"). So, here, the patentee would only collect 25% of the \$1 million in profits attributable to the patented invention.

170. Until the Federal Circuit's recent decision in *Uniloc USA, Inc. v. Microsoft Corp.*, 632 F.3d 1292 (Fed. Cir. 2011), patentees were generally entitled to 25% of the accused infringer's profits as a "rule of thumb" in reasonable royalty determinations, making that figure today more of a ceiling than a floor. See *id.* at 1312–18 (holding that evidence relying on the 25% rule of thumb is generally inadmissible).

171. John M. Golden, Commentary, "Patent Trolls" and Patent Remedies, 85 TEXAS L. REV. 2111, 2117 (2007); see also Golden, *supra* note 65, at 512 (advocating for an "antidiscrimination" principle in patent law remedies).

again fall back on the flawed conceptual tort law model of patent damages: “There is no unjustified discrimination here; in both cases, the goal of patent remedies is properly to ensure that patent owners are compensated for any unauthorized uses made by others.”¹⁷²

In their reply, Lemley and Shapiro further contend that:

While injunctions in lost-profits cases may also cause holdup problems, we do not see a practical and general way of avoiding these problems while providing adequate compensation to patent holders for infringement. There is a fundamental difference between cases involving reasonable royalties and those involving lost profits. In reasonable-royalty cases, the joint profits of the patent holder and the infringing firm are *increased* by the infringing firm’s use of the patented invention, so the hypothetical ex ante negotiation concept makes good economic sense. In lost-profits cases, the joint profits of the patent holder and the infringing firm may well be *decreased* by the infringing firm’s use of the patented invention. In such cases, the hypothetical ex ante negotiations concept is not helpful for assessing damages and it is not possible to find licensing terms that are commercially acceptable to the downstream firm and adequately compensate the patent holder.¹⁷³

Although Lemley and Shapiro are correct that in lost-profits cases, joint profits can be decreased by the infringing firm’s use, one need not rely on a hypothetical *ex ante* negotiation to calculate the proper level of ongoing damages. Rather, as in the simple numerical examples presented earlier, the question becomes what measure of monetary award would properly reward the patentee’s invention in an incentive-based model. Because, as Lemley has recognized elsewhere, a patentee need not reap the entire social value of its invention in order to be sufficiently incentivized, it is not necessary to imagine a hypothetical set of “licensing terms that are commercially acceptable . . . and [that] adequately compensate the patent holder.”¹⁷⁴ Rather, the court merely needs to award the patentee some portion of the available profit stemming from infringement adequate to *incentivize the invention*.¹⁷⁵

172. Lemley & Shapiro, Reply, *Patent Holdup and Royalty Stacking*, 85 TEXAS L. REV. 2163, 2171 (2007). In a later article Lemley argues that compensation for nonpracticing entities “should be based on the value that the patented invention actually contributes as a proportion of the defendant’s product, taking into account the other patents, know-how, raw materials, and labor that also contribute to the value of that product and the existence of possible alternatives to the patented technology.” Lemley, *supra* note 6, at 670. This more nuanced view of compensation should be applied with equal force to practicing entities.

173. Lemley & Shapiro, *supra* note 172, at 2173 n.49 (emphasis omitted).

174. *Id.*

175. Below, I advocate also considering incentives to commercialize. See *infra* notes 196–200, 240 and accompanying text.

Last, returning to their original article, Lemley and Shapiro try to escape the debate by conjecturing that ongoing lost profits are difficult to calculate.¹⁷⁶ Yet, the same kinds of difficulties in calculating and awarding lost profits on an ongoing basis beset calculating ongoing reasonable royalties.¹⁷⁷ Indeed, in one of his later articles, Lemley argues that calculating lost profits is generally more straightforward than calculating reasonable royalties:

But where the problem is imprecision in calculating lost profits, it is important to keep in mind that the alternative to denying lost profits is a less precise, and more distorted, reasonable royalty measure. . . . But under a strict divide approach, a patentee who can show that it is more likely than not that an infringer's sales cut into its own should be entitled to the court's best estimate of the patentee's lost profits. That estimate may not be perfect, but it is likely to be at least as accurate as the alternative reasonable royalty measure

Indeed, it is somewhat ironic that courts have insisted on strict compliance with the elements of proof of a lost profits claim, given that the reasonable royalty alternative involves at least as much uncertainty and approximation.¹⁷⁸

Like ongoing reasonable royalties, courts could assume that ongoing lost profits generally mimic historical profits, subject to adjustment for increased or decreased future sales.¹⁷⁹ As such, Lemley and Shapiro's claim that ongoing lost profits are too difficult to calculate is dubious. In any event, patent law should not aim to award a manufacturing patentee lost profits on an ongoing basis. Rather, the core question is what level of damages should be awarded to sufficiently induce the patentee to engage in

176. Lemley & Shapiro, *supra* note 19 ("The presumptive right to a permanent injunction in these cases is justified in part for reasons of equity and in part because of the grave difficulties associated with calculating and awarding lost profits on an ongoing basis.").

177. See Paul M. Janicke, *Implementing the "Adequate Remedy at Law" for Ongoing Patent Infringement After eBay v. MercExchange*, 51 IDEA 163, 166–67 (2011) ("[I]n the course of straying away from damages precedents and into the realm of judicially forced periodic payouts, the courts have brought about a number of difficulties for themselves, for the litigants, and for the patent system. . . . The mandatory ongoing royalty approach thus weakens the patent system as a whole, at a time when it is viewed, at least by some observers, as critical to the nation's economic wellbeing."); Stephen M. Ullmer, Note, *Paice Yourself: A Basic Framework for Ongoing Royalty Determinations in Patent Law*, 24 BERKELEY TECH. L.J. 75, 98 (2009) ("The Federal Circuit's decisions in *Paice* and *Amado* show the uncertainty involved in an ongoing royalty determination. Though the court in each case acknowledged the potential validity of an ongoing royalty, neither decision set forth a comprehensive framework for the lower courts to follow.").

178. Lemley, *supra* note 6, at 672 & n.79 (footnote omitted).

179. See Mark A. Lemley, *The Ongoing Confusion over Ongoing Royalties*, 76 MO. L. REV. 695, 701–02 (2011) ("[A] court setting an ongoing royalty after a finding of infringement is not writing on a blank slate. A jury has already set a reasonable royalty for past damages."); Ullmer, *supra* note 177, at 95 ("The reasonable royalty should act as a baseline value for the courts, which can then alter the reasonable royalty in accordance with the other essential *Georgia-Pacific* considerations.").

innovation. On this ground, as I explain more fully below, courts need not look solely to tort-based compensation—that is, lost profits or reasonable remedies—in fashioning an appropriate measure of damages.¹⁸⁰

In sum, the premise of promoting the *status quo ante* should be jettisoned as a fundamental tenet of patent law remedies. Neither nonpracticing nor manufacturing entities should routinely be entitled to an injunction as a matter of equity, other than in situations for which the costs of determining damages (including error costs from wrong determinations) exceed the costs of granting the injunction (an issue I address in more detail in the next subpart).¹⁸¹ Indeed, on the same reasoning, patentees should *not* conclusively be entitled to damages sufficient to fully compensate for the “harm” inflicted on the patentee by the infringement. Instead, patentees should only be entitled to the level of damages that promotes optimal innovation incentives, which—as I explain in the next subpart—may often be less, and sometimes more, than the profits or royalties that would have been earned in the marketplace but for the infringement.

C. Fully Rejecting Tort Law Remedies in Patent Law

In this subpart, I extend my argument that tort law principles should be excised from patent law remedies. As I recounted earlier, tort law concepts immediately lead to a distinction in the types of remedies that are awarded to manufacturing and nonpracticing entities that is not always economically justified. In this regard, tort law concepts have caused courts and commentators to improperly overlook holdup problems caused when injunctions are granted to practicing entities asserting patents on components of complex products and switching costs are high. In this subpart, I contend that injunctions and full compensatory damages may also thwart optimal innovation incentives when they generate large consumer deadweight losses, result in substantial duplicated costs during the pre-invention R&D process, or create transaction costs far in excess of the value of the invention. On the other hand, patentees may be systematically undercompensated by mere make-whole damages when infringement is difficult to detect or when private incentives to innovate from make-whole damages are insufficient to generate optimal levels of social benefits from the innovation.¹⁸²

180. See *infra* subpart II(C).

181. See Golden, *supra* note 65, at 512 (“[D]etailed tailoring of patent remedies must have limits and is only justified insofar as it promises benefits that justify its costs.”).

182. In what follows, I limit my discussion to standard modes of injunctive and monetary relief—namely, granting (or denying) injunctive relief and awarding (or not) some measure of monetary damages to the patentee. In a recent article, Dan Burk offers an original, nearly dizzying, array of nonstandard remedies that revolve around various “real options” that the court can award either to the patentee or the accused infringer. Dan Burk, *Intellectual Property in the Cathedral*, in ACCESS TO INFORMATION AND KNOWLEDGE: 21ST CENTURY CHALLENGES IN

1. Correcting Patentee Overcompensation.—In the last subpart, I argued that injunctions involving patents covering components of complex products should typically be denied—at least when switching costs are high—not only to nonpracticing entities, as commentators have recognized, but also to practicing entities. In this section, I examine other areas in which make-whole damages may overcompensate patentees.

First, for reasons similar to those presented earlier for denying injunctions for component patents, overall damages for practicing entities should not always reflect the entirety of the patentee's lost profits attributable to the infringement. Suppose the patentee sells a product for \$100 with \$2 of marginal profit per product, which reflects the minimum profit necessary for competitors to stay in the product market. In other words, competitors could earn slightly less than 2% profit margins by making other products. Again, suppose that the invention allows the patentee to reduce its costs of manufacturing a component of the product by \$1. If competitors cannot use the patented cost-saving method or a suitable design-around with similar cost-savings, the patentee could lower its price to just below \$100, potentially forcing its competitors out of the product market.

In the event of infringement by a competitor, under the “entire market value” rule of current law, the patentee would be entitled to full lost profits because the competitors’ customers would have bought the patentee’s product in the absence of infringement precisely because of its patented production method and resulting cheaper component, which is the only reason for the lower price of the product.¹⁸³ If nine competitors each sell

INTELLECTUAL PROPERTY AND KNOWLEDGE GOVERNANCE 95 (Dani Beldiman ed., 2013). For instance, Burk describes a sort of “put” rule under which “the alleged infringer continues to use the intellectual property, but can choose to stop infringing and receive damages from the intellectual property owner.” *Id.* at 106. Such a rule potentially presents a more beneficial alternative than merely denying injunctive relief or awarding less than lost profits. However, in order to simplify the exposition here, I leave such considerations for future work.

183. For example, the Federal Circuit upheld a finding of lost profits on patents reducing the cost of producing VCRs under the entire market value rule because “[t]he evidence at trial portrayed general industry demand for smaller, cheaper, faster, and more reliable VCRs, and [the patentee] presented evidence that the patented technology furthers these goals.” *Funai Elec. Co., v. Daewoo Elecs. Corp.*, 616 F.3d 1357, 1362, 1375–76 (Fed. Cir. 2010) (“The patented inventions are described as improvements that lower the cost of producing VCRs while maintaining product quality.”); *see also* *Garretson v. Clark*, 111 U.S. 120, 121 (1884) (“The patentee . . . must show . . . that the profits and damages are to be calculated on the whole machine, for the reason that the entire value of the whole machine, as a marketable article, is properly and legally attributable to the patented feature.” (internal quotation marks omitted)); *LaserDynamics, Inc. v. Quanta Computer, Inc.*, 694 F.3d 51, 67 (Fed. Cir. 2012) (“If it can be shown that the patented feature drives the demand for an entire multi-component product, a patentee may be awarded damages as a percentage of revenues or profits attributable to the entire product.”); *TWM Mfg. Co. v. Dura Corp.*, 789 F.2d 895, 901 (Fed. Cir. 1986) (“The entire market value rule allows for the recovery of damages based on the value of an entire apparatus containing several features, when the feature patented constitutes the basis for customer demand.”).

100,000 products per year prior to the patented invention, assuming for simplicity that all competitors infringed, the patentee would garner just less than \$300,000 per year (that is, just less than \$3 per product, the original \$2 in profit plus the extra \$1 afforded by the patent) in damages from each infringing competitor, even though the value of the infringing component was \$100,000 per year (that is, \$1 per product) for each competitor. This excessive level of damages, which is triple the social value of the invention, is only justifiable on tort law grounds as a matter of compensation for the “harm” caused by infringement. Like the arguments in favor of apportionment for reasonable royalty damages,¹⁸⁴ full lost profits will often be inappropriate for patents merely covering components of more complex products.

Second, in many contexts besides component patents, the current system of patent remedies can generate excessive innovation incentives. This is especially so because R&D and commercialization costs and risks are substantially more in some industries than others. For instance, the costs of invention in the software industry are usually far below those in the pharmaceutical industry.¹⁸⁵ Yet, the duration of software and pharmaceutical patents are exactly the same (in practice, software patents last longer), and the effective scope of software patents often exceeds the scope of pharmaceutical patents.¹⁸⁶ If the broad scope and long duration afforded software patents is unnecessary to fully incentivize innovation in that industry, then the rewards provided by the patent system are excessive.¹⁸⁷ This is so even if the social value of an invention far exceeds the R&D costs necessary to produce it because society should only pay the inventor the minimum amount necessary to generate the innovation.¹⁸⁸ As such, paying full damages in situations where R&D and commercialization costs and risks are low creates windfalls for innovators.¹⁸⁹

184. See Lemley & Shapiro, *supra* note 19, at 2023–24 (suggesting that reasonable royalty damages should be solely based on the value added by the patented component and that total damages should be accordingly apportioned).

185. See Burk & Lemley, *supra* note 26, at 1581–83 (noting that R&D costs for software are considerably lower than those for pharmaceuticals).

186. See Morris, *supra* note 27 (noting that the effective patent term for pharmaceuticals, even with patent term restoration under the Hatch-Waxman Act, is fourteen years while the effective term for nonpharmaceuticals patents is eighteen and a half years).

187. See Burk & Lemley, *supra* note 26, at 1687 (“While most biotechnological and chemical inventions require broad patent protection because of their high cost and uncertain development process, the opposite is true in the case of software development. Software inventions tend to have a quick, cheap, and fairly straightforward post-invention development cycle.”).

188. See Lemley, *supra* note 25, at 1031 (arguing that the fundamental goal of patent law is “to give as little protection as possible consistent with encouraging innovation”).

189. See Cohen & Noll, *supra* note 29, at 458 (“[I]f IP protection is too strong, it can reduce economic welfare by inhibiting some kinds of technological change and excessively encouraging other types relative to the progress that would be forthcoming with weaker IP protection.”); cf. Tun-Jen Chiang, *A Cost-Benefit Approach to Patent Obviousness*, 82 ST. JOHN’S L. REV. 39, 73

Additionally, awarding injunctive relief and full damages can create deadweight losses that stem from (1) the patentee's ability to price its patented goods far over the competitive price;¹⁹⁰ and (2) excessive rents generated from duplicated R&D in "racing" for too large a patent prize.¹⁹¹ Thus, when R&D costs and risks are low and deadweight losses and rent-seeking costs are high, injunctions should be routinely denied and damages reduced when the potential error costs of doing so are low.¹⁹² Indeed, Ian Ayres and Paul Klemperer¹⁹³—as well as one of my papers¹⁹⁴—have shown that allowing for imperfect enforcement of the patent can result in superior welfare outcomes to full enforcement, particularly under these conditions.¹⁹⁵

Third, in commercial situations involving large transaction costs in negotiating licenses, traditional injunctive relief may diminish the commercialization and improvement of patented inventions.¹⁹⁶ Specifically, when a patentee and a third-party commercializer or improver substantially differ over whether a given patent is infringed, valid, or enforceable, it may be economically efficient for the third party to forgo large transaction costs in negotiating a license, instead infringing. Like the theory of efficient breach in contract law,¹⁹⁷ efficient infringement occurs when the transaction costs of negotiation dwarf the value of the innovation at issue, which can occur when there is large uncertainty in the underlying patent rights,¹⁹⁸ or simply when the economic value of the innovation is

(2008) ("[P]atents may create such enormous incentives that research funds are diverted from other places where they could have been more socially efficiently invested . . .").

190. See Lemley, *supra* note 25, at 1059 ("By definition . . . the intellectual property system permits owners to raise price above marginal cost, creating deadweight losses by raising the price to consumers.").

191. See Grady & Alexander, *supra* note 31 ("The defect in the system is that if multiple inventors expend resources in competition for the patent monopoly, the benefit to society of having the invention will be dissipated by the cost of numerous, redundant, development efforts.").

192. Cf. Lederman & Sichelman, *supra* note 42, at 1685 (proposing a measured enforcement approach whereby a tax enforcer intentionally engages in imperfect enforcement so as to reduce deadweight losses).

193. See Ayres & Klemperer, *supra* note 46, at 993.

194. See Sichelman, *supra* note 47.

195. Reducing overall damages can be achieved not only through reductions on damages for sales of products covered by the patent but also by limiting damages on "convoyed" sales of related products. See Lemley, *supra* note 6, at 660 (discussing convoyed sales jurisprudence).

196. See Sichelman, *supra* note 35, at 384 ("Like the ex ante licensing context . . . strong, property-like rights will exacerbate the effects of strategic, ex post litigation by increasing the risk of infringement, making the damage award or injunction value to the patentee inefficiently high relative to the social value of commercialization.").

197. See RICHARD A. POSNER, *ECONOMIC ANALYSIS OF LAW* 119–21 (7th ed. 2007); Macneil, *supra* note 33.

198. See Depoorter, *supra* note 8, at 67–70 (describing how "innovation uncertainty" can frustrate licensing deals).

fairly minimal.¹⁹⁹ In these circumstances, it may be more optimal to preclude injunctions, even for practicing patentees.²⁰⁰

2. *Correcting Patentee Undercompensation.*—Eliminating the presumption of tort law that a patentee should be made whole need not merely lead to eliminating injunctions and reducing overall compensation. Here, I describe two situations in which the patentee may need *more than* injunctive relief and compensatory damages to generate optimal innovation incentives.

First, infringement is often hard to detect.²⁰¹ For instance, if a competitor uses a patented process in a manufacturing process, it may be very difficult, even impossible, to determine whether a given product was made by the process.²⁰² Even in the make-whole realm of tort law, when violations are not easily detected, standard law-and-economics analysis mandates that for optimal deterrence, damages must be multiplied by the inverse of the probability of detection.²⁰³ Thus, if detection of infringement is difficult, damages should be enhanced to provide optimal innovation incentives.²⁰⁴ As Roger Blair and Thomas Cotter have recognized, “an award that merely renders the infringer no better off as a result of the infringement may be an ineffective deterrent, because only a portion of all possible infringements are susceptible of detection. This insight suggests that a substantial damages multiplier often may be necessary to achieve adequate deterrence.”²⁰⁵ Because detection may be difficult irrespective of

199. Cf. Jeffrey L. Harrison, *A Positive Externalities Approach to Copyright Law: Theory and Application*, 13 J. INTELL. PROP. L. 1, 49 (2005) (“When transaction costs are high and the fair user attributes greater value to the use of the work than the loss suffered by the original author, it can be efficient to permit the fair use.”).

200. Cf. Sichelman, *supra* note 35 (advocating that reducing effective remedies for the patentee may be efficient when the patentee fails to commercialize the patented invention); Katherine J. Strandburg, *Patent Fair Use 2.0*, 1 U.C. IRVINE L. REV. 265, 299–300 (2011) (proposing a fair use defense when, *inter alia*, transaction costs prevent a failure to license the patented technology); Julie S. Turner, *The Nonmanufacturing Patent Owner: Toward A Theory Of Efficient Infringement*, 86 CALIF. L. REV. 179, 204 (1998) (advocating “efficient infringement[]” solely for “unused” patents).

201. See Stuart J.H. Graham & Ted Sichelman, *Why Do Start-Ups Patent?*, 23 BERKELEY TECH. L.J. 1063, 1086 n.101 (2008) (“[T]he ability to detect infringement in the first instance . . . may be very costly itself . . .”).

202. See, e.g., Jochen Pagenberg, *The WIPO Patent Harmonization Treaty*, 19 AIPLA Q.J. 1, 14 (1991) (noting the difficulty of detecting process patent infringement).

203. See Richard Craswell, *Deterrence and Damages: The Multiplier Principle and Its Alternatives*, 97 MICH. L. REV. 2185, 2186 (1999) (“[T]he ideal penalty (insofar as deterrence is concerned) equals the harm caused by the violation multiplied by one over the probability of punishment.”). See generally Gary S. Becker, *Crime and Punishment: An Economic Approach*, 76 J. POL. ECON. 169 (1968) (describing the theory of punishment multipliers).

204. See Heald, *supra* note 19, at 1182 (“[O]ne can justify an award of augmented damages in order to account for the likelihood that the infringement would go undetected.”).

205. See Blair & Cotter, *supra* note 37.

the infringer's behavior, this condition applies regardless of whether the infringer was willful, unless the willfulness somehow increases the odds of nondetection.

A potential counterargument is that courts will themselves have difficulty determining when infringement is difficult to detect. One possible administrative solution is for the courts to assess whether the accused product or method was sold or used in public. If not, then courts should presume, rebuttably, that the accused product or method is difficult to detect. Assuming the presumption holds, then the question becomes, "How difficult?" This is an even thornier inquiry. Interestingly, as Paul Heald has recognized, the oft-discredited "disgorgement" remedy—whereby the accused infringer is required to pay no less than the profits it earned from infringement—may become an effective remedy because it achieves some multiplier on ordinary damages, yet is much less costly to administer than an exact-multiplier remedy.²⁰⁶ Of course, it would be ideal for courts to fashion a low cost remedy that roughly determined the appropriate multiplier.

Second, make-whole damages may sometimes be too low to generate sufficient positive incentives for any private actor to invest in R&D and commercialization for a given invention, but the net social welfare of the invention may be sufficiently high that society desires the invention to be made and commercialized.²⁰⁷ Alternatively, the invention may be justified purely on distributive grounds, such as drugs for rare diseases that are not otherwise profitable.²⁰⁸ In these situations, infringers, the government, or some other entity would need to supplement make-whole damages with additional funds.

Presumably, other than the possibility of disgorgement of profits, infringers should not pay these additional amounts because assessing such liability would lead to allocative distortions in which potential infringers take excessive caution.²⁰⁹ As I explained earlier, in many circumstances, infringement can be a kind of efficient breach that diminishes the deadweight losses imposed by patents.²¹⁰ In these cases, if we raise the costs of infringement, we lower the benefits from otherwise efficient

206. See Heald, *supra* note 19, at 1197–98 (“[I]t suggests the infringer’s profits should sometimes be part of a remedy for patent infringement (as it is for trademark and copyright infringement), supporting arguments that it should be reinstated.”).

207. Cf. Shavell & Ypersele, *supra* note 38 (“[I]ncentives to invest in research are inadequate because monopoly profits are less than the social surplus created by an innovation.”).

208. See Sichelman, *supra* note 35, at 385–88 (discussing distributive values in patent law).

209. See Daniel A. Crane, *Optimizing Private Antitrust Enforcement*, 63 VAND. L. REV. 675, 715 (2010) (“For example, granting an award of treble damages . . . entails a very serious risk of chilling . . . innovation, one of the very things that both antitrust and patent law seek to stimulate.”).

210. See *supra* notes 196–200 and accompanying text.

infringement. Rather, a more appropriate approach would be for the government to step in and determine whether current R&D and commercialization incentives, which stem in part from patent remedies, are sufficient for particular kinds of inventive activity. If incentives are too low, the government could—as it does now—subsidize certain categories of inventive activity (e.g., drugs for rare diseases, assistive devices for the disabled, and renewable-energy technologies) both prior to and after invention.²¹¹

III. Some Possible Costs and Criticisms of Judicially Measured Remedies

In arguing in favor of measuring remedies so as to provide more or less to the patentee than make-whole remedies, I generally ignored several potential costs and criticisms of my proposal. First, as I explained, changing remedies may effectively alter the underlying substance of the patent laws—not only in terms of incentives provided to inventors and commercializers but also to potential infringers. To the extent judges ascribe to themselves a power that may effectively contravene statutory or regulatory law, one might argue that doing so impermissibly treads on constitutional separation of powers principles and related democratic concerns. Second, deviating from the baseline of make-whole damages may increase overall uncertainty, potentially deterring future innovation. Relatedly, juries and judges may not have the ability to accurately award damages and determine injunctive relief other than to make plaintiffs whole. In this event, moving to a system of measured remedies—whereby judges and juries have significant discretion to depart from the tort law baseline—would merely exacerbate present errors, leading to inferior results. Here, I address these concerns in turn.

A. *Contravening Democracy and the Separation of Powers*

Because the current patent-remedies scheme contemplates making the patentee whole in the event of infringement, attempts to deviate from that baseline—even for awarding reasonable royalties or equitable remedies²¹²—would contravene legislative authority.²¹³ Instead, as I argue

211. See, e.g., Orphan Drug Act, Pub. L. No. 97-414, 96 Stat. 2049 (1983) (codified as amended in scattered sections of 21, 26 & 42 U.S.C.) (providing short-term, seven-year market-exclusivity rights similar to patent rights; subsidies for clinical trials; tax incentives; and exemption from certain FDA registration fees for companies developing drugs for rare diseases).

212. Recently, in *Apple v. Motorola*, Judge Posner denied Apple injunctive relief, in part because:

[I]t would be likely to impose costs on the alleged infringer disproportionate both to the benefits to it of having infringed and to the harm to the victim of infringement, and would thus be a windfall to the patentee and a form of punitive rather than compensatory damages imposed on the infringer. . . . [A]n injunction could force Motorola to remove lucrative products from the market for as long as it took to remove the infringing features—minor features in complex devices most features of

below, modification of the existing statutory framework by Congress would be essential to carrying out my proposal. Even still, would congressionally authorized judicial discretion somehow compromise separation of powers or democratic principles?²¹⁴ Specifically, because Congress is slated with “making” the law and the courts with “interpreting” the law, one might argue that judicially “dialing up” or “dialing down” remedies on the basis of the economic factors perverts the notion of substantive law to such a great extent that it unconstitutionally impedes on Congress’s ability to legislate.²¹⁵ Although such a view may have some force, the same could be said of executive discretion to enforce the law and judicial discretion to fashion remedies in other areas of the law (e.g., criminal law). Any broad

which are not alleged to infringe—from its products, or to invent around the infringing features.

See Apple, Inc. v. Motorola, Inc., 869 F. Supp. 2d 901, 917 (N.D. Ill. 2012). Although Judge Posner’s actions are consistent with the normative views presented here, they are unlikely to pass judicial muster on appeal. Specifically, despite Judge Posner’s argument that the disproportionate relief that would be enjoyed from an injunction shows the absence of irreparable harm, *see id.*, generally the courts have interpreted irreparable harm in a measure consistent with restoring the *status quo ante*, *see supra* Part I. Because Apple practices the patent-in-issue—and, unlike Motorola, did not make any *ex ante* licensing commitments, *see supra* note 107—the *status quo ante* here would bar the practicing of Apple’s patent by Motorola. Thus, under the present case law—which is consistent with Congress’s intent—there would be no irreparable harm, despite the economic windfall to Apple. For instance, in *Robert Bosch LLC v. Pylon Manufacturing Corp.*, 659 F.3d 1142 (Fed. Cir. 2011), the Federal Circuit reversed a denial of injunctive relief, relying in part on the facts that the parties were direct competitors and the patentee faced loss of market share. *Id.* at 1152–54. Although the Federal Circuit affirmed a denial of a preliminary injunction in *Apple v. Samsung*, the court found that “Apple ha[d] presented no evidence that directly ties consumer demand for the [infringing product] to its allegedly infringing feature.” *Apple Inc. v. Samsung Elecs. Co.* 695 F.3d 1370, 1375 (Fed. Cir. 2012). Thus, in *Apple v. Samsung*, no injunction was necessary to restore the *status quo ante* because on the record essentially the same number of infringing products would have been sold even if the product had not incorporated the infringing component. *See id.* at 1377 (holding that no irreparable harm occurred because Apple failed to establish a causal nexus between the allegedly infringing features and consumer demand); *see also Apple Inc. v. Samsung Elecs. Co.*, 735 F.3d 1352, 1364–65, 1375 (Fed. Cir. 2013) (remanding for further findings regarding customer demand in the context of a permanent injunction determination). Thus, there seems little support in Supreme Court or Federal Circuit case law for Judge Posner’s normatively admirable, but judicially questionable, approach.

213. *See, e.g., Gen. Motors Corp. v. Devex Corp.*, 461 U.S. 648, 653–55 (1983) (“When Congress wished to limit an element of recovery in a patent infringement action, it said so explicitly. . . . Congress’ overriding purpose [is] affording patent owners complete compensation.”).

214. *See generally* Jack M. Beerman, *An Inductive Understanding of Separation of Powers*, 63 ADMIN. L. REV. 467, 472–81 (2011) (describing separation of powers principles).

215. *See* Lederman & Sichelman, *supra* note 42, at 1739 n.268 (“In theory, a measured enforcement regime, particularly in the federal context, may raise concerns that it could override congressional intent, violate separation of powers, or violate due process requirements.”); *see also* BLAIR & COTTER, *supra* note 9, at 6 (“More fundamentally . . . this view threatens to undermine the rule of law by substituting judges’ idiosyncratic views of the merits of (say) the patent system for the view expressed by Congress in enacting the Patent Act.”); *id.* (“We believe that, as a general matter, courts *ought* to operate as if the intellectual property laws embody the proper balance when deciding how to best calculate damages . . . and leave it to other branches of government to decide whether the underlying assumption is true or false.”).

discretion can drastically affect the incentives for legal actors to engage in or refrain from particular behavior and, in turn, substantially affect the economic sphere.²¹⁶ Discretion in patent law remedymaking seems no different in kind from discretion in other areas of the law, which have existed before and after the time of the Constitution.²¹⁷ As such, as long as Congress provides the courts with such discretion, separation-of-powers and democratic concerns seem satisfactorily addressed.²¹⁸

B. *The Costs of Uncertainty*

Of course, with greater discretion comes greater uncertainty. In general, legal uncertainty can create significant public and private costs in complying with the law and bargaining in its shadow.²¹⁹ Yet, uncertainty is relative and the costs it creates must be measured against the costs avoided by affording courts the leeway to fashion economically sensible remedies. Although the current system revolves around a baseline of making patentees whole in the event of the infringement, awarding remedies is anything but an exact science.²²⁰ Deviating from current damages calculations to take into account the economic incentives necessary to motivate invention and innovation—and to avoid duplicated costs and deadweight losses—seems unlikely to generate significant additional uncertainty in the long term. This is especially so if such deviation is limited to cases in which make-whole remedies are clearly economically inferior, as I advocate in the following subpart.

Of greater concern is providing courts even more leeway to deny injunctions in favor of ongoing monetary damages. As I explained, *eBay* eliminated the nearly irrebuttable presumption of injunctive relief.²²¹ Scott

216. Cf. Lederman & Sichelman, *supra* note 42, at 1739–40 (“The reality is that agencies already have enforcement discretion.”).

217. See Emily Sherwin, *Introduction: Property Rules as Remedies*, 106 YALE L.J. 2083, 2085 (1997) (“[T]he great body of property rules are enforced by courts through equitable remedies such as injunctions and specific performance—remedies that historically have been surrounded by a cloud of judicial discretion.”); Michael Barone, Jr., Comment, *Delegation and the Destruction of American Liberties: The Affordable Care Act and the Contraception Mandate*, 29 TOURO L. REV. 795, 804 n.72 (2013) (“The Court has a long history of allowing Congress to grant Executive agencies broad discretion in determining if Congress’s vague standards have been met.”).

218. See, e.g., *Mistretta v. United States*, 488 U.S. 361, 412 (1989) (affirming judicial discretion in criminal sentencing); *Wagner Seed Co. v. Daggett*, 800 F.2d 310, 316 (2d Cir. 1986) (“[T]here is no constitutional violation if the imposition of penalties is subject to judicial discretion . . .”).

219. See Richard E. Levy & Robert L. Glicksman, *Agency-Specific Precedents*, 89 TEXAS L. REV. 499, 574 (2011) (“For private parties, legal uncertainty increases information costs, requires additional planning, and creates risk. . . . For courts, legal uncertainty leads to litigation and makes settlement more difficult because parties may entertain substantially different assessments of the likely outcome of litigation.”).

220. See *supra* notes 176–180 and accompanying text.

221. See *supra* notes 11–19 and accompanying text.

Kieff, Henry Smith, Richard Epstein, and others have stringently criticized this move as introducing substantial uncertainty, as well as increasing potential error costs, thereby decreasing incentives to strike deals outside litigation.²²² Although I agree these costs may be large, the costs avoided in denying economically unjustified injunctions are likely to be even larger in many situations.²²³ Additionally, although private parties may sometimes be able to contract around inefficient injunctive remedies,²²⁴ high bargaining costs and information asymmetries will often prevent such deals, leaving a substantial residuum of inefficient cases.²²⁵ Ultimately, the relative balancing of costs in these cases will turn on the ability of judges to

222. See F. Scott Kieff & Henry E. Smith, *How Not to Invent a Patent Crisis*, in REACTING TO THE SPENDING SPREE: POLICY CHANGES WE CAN AFFORD 55, 67 (Terry L. Anderson & Richard Sousa eds., 2009) (“[A]n injunction forces the infringer to stop and enforces the delegation of valuation questions to patentees and their contractual partners, with a view toward markets for inputs and products, rather than officials, courts, and experts for hire.”); see also Richard A. Epstein, *The Disintegration of Intellectual Property? A Classical Liberal Response to a Premature Obituary*, 62 STAN. L. REV. 455, 488 (2010) (arguing that injunctive relief “keeps open the channels for voluntary exchange”); F. Scott Kieff, *Removing Property from Intellectual Property and (Intended?) Pernicious Impacts on Innovation and Competition*, 19 SUP. CT. ECON. REV. 25, 41 (2011) (“A central and underexplored problem with liability rules is that they seriously frustrate the ability for the patentee to attract and hold the constructive attention of a potential contracting party while preserving the option to terminate the negotiations in favor of striking a deal with a different party. This comparative effectiveness of property rules in achieving these goals is the mechanism by which property rules facilitate both innovation and competition.”); Sandrik, *supra* note 4, at 136 (“[P]roperty rules in patent law encourage parties to interact and reach a mutually beneficial bargain.”); cf. BLAIR & COTTER, *supra* note 9, at 40 (“[A] court setting the terms of the exchange would have a difficult time doing so quickly and cheaply, given the specialized nature of the assets and the varied and complex business environments in which the assets are deployed.”); Blair & Cotter, *supra* note 19, at 48 (arguing that a property rule “encourage[s] would-be users of the patented invention to negotiate with the patent owner, the theory being that the outcome of these private negotiations . . . more accurately reflects the value of the invention than would a compulsory-licensing fee set by the government”); Robert P. Merges, *Contracting into Liability Rules: Intellectual Property Rights and Collective Rights Organizations*, 84 CALIF. L. REV. 1293, 1299 (1996) [hereinafter Merges, *Contracting into Liability Rules*] (“[P]rivate, voluntary organizations . . . would be superior to state-mandated compulsory licenses.”); Robert P. Merges, *Of Property Rules, Coase, and Intellectual Property*, 94 COLUM. L. REV. 2655, 2662 (1994) [hereinafter Merges, *Of Property Rules*] (arguing that in a property rule regime, IP owners will have strong incentives to form institutions that are “designed to streamline the exchange of property rights” by modifying “the strong property rule baseline of intellectual property law by contracting into liability rules”).

223. Even Smith recognizes that in some situations, money damages may be more appropriate. See Mark P. Gergen, John M. Golden & Henry E. Smith, *The Supreme Court’s Accidental Revolution? The Test for Permanent Injunctions*, 112 COLUM. L. REV. 203, 249 (2012) (noting that denying injunctive relief can sometimes perform an important “safety valve” function that “both prevents ex post waste and, more importantly, discourages opportunism”).

224. See Merges, *Contracting into Liability Rules*, *supra* note 222 (noting that private collective rights organizations are well-attuned to the needs of specific industries and provide robust mechanisms for contracting around compulsory licenses and injunctive remedies with relatively low transaction costs).

225. Additionally, parties can also contract around inefficient liability rules. Mark A. Lemley, *Contracting Around Liability Rules*, 100 CALIF. L. REV. 463, 464 (2012).

recognize (or not) those specific situations in which denying injunctions is economically sound, to which I turn next.

C. *Institutional and Cognitive Limitations of Courts and Judges*

Clearly, courts may find it challenging to award an appropriate level of ongoing money damages, regardless of how damages are calculated,²²⁶ and there is reason to believe courts may systematically undercompensate patentholders.²²⁷ An injunction, on the other hand, effectively forces the parties to enter into a private negotiation to determine price terms—assuming, of course, that they can come to some agreement.²²⁸ More specifically, injunctions (i.e., property rules) are more optimal than damages (i.e., liability rules) when the marginal judicial costs of determining damages and the error costs from wrong determinations exceed the marginal costs of granting an injunction (relative to awarding damages).²²⁹

226. See Janicke, *supra* note 177 (arguing that numerous uncertainties are introduced under an ongoing royalties approach); Stewart E. Sterk, *Property Rules, Liability Rules, and Uncertainty About Property Rights*, 106 MICH. L. REV. 1285, 1313 (2008) (contending that a pure damages regime results in a “loss of all protection for subjective value, together with increased need for judicial assessment of damages”); Thomas F. Cotter, *The Comparative Law and Economics of Standard-Essential Patents and FRAND Royalties* 31 (Univ. of Minn. Law Sch. Legal Studies Research Paper Series, Paper No. 13-40, 2013) (“[T]he principal economic justification for awarding permanent injunctions in patent infringement cases is a matter of information costs: that is, the assumption that the parties . . . are in a better position than is a court or agency to determine what the patent is worth . . .”); cf. PHILLIP E. AREEDA & HERBERT HOVENKAMP, 3 ANTITRUST LAW: AN ANALYSIS OF ANTITRUST PRINCIPLES AND THEIR APPLICATION ¶ 707j, at 293 (3d ed. 2008) (stating that patent “value is almost impossible to determine, apart from such an obvious case as an improved process that reduces everyone’s production costs by, say, 10 percent”).

227. Blair & Cotter, *supra* note 19, at 69 (contending that reducing damages from a make-whole level could result in too little invention); Peter Camesasca et al., *Injunctions for Standard-Essential Patents: Justice is Not Blind*, 9 J. COMPETITION L. & ECON. 285, 288, 290 (2013) (arguing that injunctions protect against the risk of “reverse holdup,” namely opportunistic behavior by potential licensees to “propose rates that are significantly below the fair [market] value”); Einer Elhauge, *Do Patent Holdup and Royalty Stacking Lead to Systematically Excessive Royalties?*, 4 J. COMPETITION L. & ECON. 535, 557 (2008) (contending that hindsight bias, inaccurate damages-estimation techniques, and the difficulties involved in communicating to juries typically leads to undercompensatory damages); Richard A. Epstein, *A Clear View of The Cathedral: The Dominance of Property Rules*, 106 YALE L.J. 2091, 2093 (1997) (“The risk of undercompensation in such situations is pervasive . . .”).

228. See Merges, *Of Property Rules*, *supra* note 222, at 2664 (“Because each [asset covered by an IP Right] is in some sense unique . . . it is difficult for a court in an infringement case to properly value the right holder’s loss. Hence, the parties should be left to make their own deal.”).

229. See Mark A. Lemley & Philip J. Weiser, *Should Property or Liability Rules Govern Information?*, 85 TEXAS L. REV. 783, 840 (2007) (“Ultimately, policymakers must evaluate the comparative costs and benefits of relying on a property rule regime enforced by courts and accompanied by antitrust oversight against a liability rule regime superintended either by courts or agencies.”). The same reasoning holds when the costs of determining less than make-whole damages, as well as related error costs, exceed the costs from awarding make-whole damages.

As an empirical matter, it is difficult to measure how often judicial determination and error costs outstrip the costs from granting injunctions (and make-whole damages).²³⁰ Of course, it can be even more troublesome for a judge to determine potential error costs in any given case.²³¹ For instance, in my hypotheticals above, I assumed that a host of variables were easily measurable. In actuality, it will often be exceedingly difficult to determine even approximate private economic values of patented technologies—and associated revenues, profits, and costs—much less social values.²³²

Yet, the same kinds of evidence required to determine social values are available under today's damages regime.²³³ Using this and related evidence, courts can arguably determine at least when the social costs from injunctions and make-whole damages are substantial compared with the social benefits of the invention. For instance, courts can hear evidence on R&D, testing, and commercialization costs (including the cost of failures); technological and market risk; increased profits versus baseline profits; the value of other patented components; the value of noninfringing alternatives; and so forth, in order to determine when injunctions and make-whole damages might lead to grossly excessive awards.²³⁴ Upon such an examination, courts would routinely find without too much labor or error that—for the very reasons presented earlier—awarding injunctions and make-whole damages in certain situations can lead to drastically suboptimal outcomes.²³⁵ These scenarios include component patents involving high switching costs; high transaction cost, low value inventions; and low R&D

230. See generally Cotter, *supra* note 226, at 34–37 (explaining the variety of complex factors that play a role in determining whether “the social harm from [holdup] is greater than the social harm resulting from [reverse holdup]”).

231. Blair & Cotter, *supra* note 19, at 69 (criticizing Ayres and Klemperer's partial damages theory for presenting “daunting empirical questions” in practice).

232. See Golden, *supra* note 171, at 2115–16 (questioning whether the economic value of a patent can be precisely defined).

233. See, e.g., Roy J. Epstein, *The Market Share Rule With Price Erosion: Patent Infringement Lost Profits Damages After Crystal*, 31 AIPLA Q.J. 1, 32–33 (2003) (recounting the types of “complex evidence” reviewed in a particular patent case, including “[the patentee's] market share, volume discount policy, strategy of dropping the [product's] effective price by adding . . . features but not increasing prices; [the accused infringer's] market share, sales, profits, growth, and selling costs; the level of prices and competition for different product models; entry by new competitors; and expert testimony”).

234. For a more comprehensive list of the types of relevant evidence courts should consider, see *infra* subpart IV(A).

235. Ben Depoorter has aptly noted:

[P]roperty-rights oriented scholars have traditionally presumed that the transaction costs involved in patent license negotiations are negligible compared with the information costs involved when courts apply compulsory licenses (liability rule protection) to patents. Accordingly, property rules provide better footing for consensual agreements in the area of patents, without having courts impose prices on innovation. This conventional wisdom is outdated.

Depoorter, *supra* note 8, at 66.

cost, low commercialization cost, low risk industries.²³⁶ At least in these situations, patent law remedies typically should be modified to deny injunctions and reduce overall damages. In closer cases, especially given the historical practice—and more in line with Kieff and Smith's suggestion—a default rule of injunctive relief and make-whole damages seems more appropriate, at least until we are quite confident in the abilities of adjudicators.

IV. Patent Law as a Public Regulatory Mechanism

In this last Part of the Article, I argue that the previous analysis implies that the current structure of remedies under the Patent Act is fundamentally flawed because it rests on a make-whole notion of damages that often fails to optimally promote innovation. Instead, I suggest that patent law remedies should be viewed not as a means to remedy private wrongs but as part of a larger regulatory mechanism grounded in public law concepts.

A. *The Fundamentally Flawed Foundation of the Remedies Provisions in the Patent Act*

In Part II, I showed that make-whole damages can often be too high or too low to incentivize optimal levels of innovation. As I explained in Part I, the reason courts impose make-whole damages is because they are mandated to do so by the Patent Act itself. First, 35 U.S.C. § 284 states that “[u]pon finding for the claimant the court shall award the claimant damages

236. Kieff and Smith criticize the view that “[t]he ideal benchmark would be an all-knowing planner who would hand out checks in the exact minimum amount to induce the invention in question—and in any event no greater than the invention’s social value” on the ground that it ignores the coordinating function of patents, which ultimately leads to commercialization. Kieff & Smith, *supra* note 222, at 62 (“Patents help achieve this socially constructive coordination by allowing . . . various actors to interconnect with each other like modules of a larger system.”). Although Kieff and Smith are correct to include the larger aim of commercialization within patent law’s ambit, liability rules need not necessarily lead to less commercialization. Indeed, injunctive remedies not only lead to beneficial coordination but also to detrimental transaction costs that can prevent bargaining among commercial actors, especially for products involving multiple components. See Sichelman, *supra* note 35, at 382–88 (explaining the varieties of transaction costs resulting from overly strong and broad patent protection); Bell & Parchomovsky, *supra* note 73, at 590 (suggesting that liability rules may be useful in overcoming strategic obstacles to successful negotiations); Depoorter, *supra* note 8 (arguing that “boundary costs, fragmentation costs, and the costs of bundling necessities” that arise from property rules “have substantially increased the burden on patent licensing markets”); see also Ian Ayres & Eric Talley, *Solomonic Bargaining: Dividing a Legal Entitlement to Facilitate Coasean Trade*, 104 YALE L.J. 1027, 1036–72 (1995) (suggesting that liability rules tend to induce actors to disclose more information otherwise withheld under a property-rule regime, increasing the odds that deals are consummated). Thus, Kieff and Smith’s claim that “the fear of weak enforcement creates a disincentive for the necessary parties to work together at the outset” is not a universal fact. Kieff & Smith, *supra* note 222, at 63. Rather, the truth lies somewhere in the middle, and it will vary with the facts in each case—facts that an adjudicator can, and should, take into account in fashioning an appropriate remedy.

adequate to compensate for the infringement but in no event less than a reasonable royalty for the use made of the invention by the infringer, together with interest and costs as fixed by the court.”²³⁷ In other words, the Patent Act enshrines the tort law compensatory rationale right into the statutory framework for damages.²³⁸

Based on the analysis I provided earlier, such an approach is misguided because it does not provide for optimal remedies in many circumstances. Rather, the Patent Act would encourage more optimal levels of innovation by giving the court discretion to adjust damages upward or downward from make-whole compensatory levels, particularly when (1) the novel aspect of the invention is but a mere component of a complex invention and switching costs are high; (2) full compensatory damages substantially exceed or fall below the amount needed to incentivize research, development, and commercialization costs and risks of the invention, including opportunity costs; and (3) infringement is especially difficult to detect.²³⁹ Of course, there are very likely many other situations that warrant damages adjustment; ultimately, a more general approach that determined remedies on the basis of innovation incentives per se would be ideal.

As I outlined earlier, such a test would take into account a variety of factors, such as total R&D costs, testing costs (including regulatory approval), commercialization costs, increased profits versus baseline profits, comparable royalties, the value of other patented components, invention risk, commercialization risk, competition risk (including the risk of free riding), available profit from use or licensing, potential R&D duplication costs, potential consumer deadweight losses, the social benefit of the invention relative to noninfringing alternatives, the ease of detecting infringement, other barriers to entry (including other forms of IP), and the like in order to appropriately gauge remedies for infringement.²⁴⁰ Formulating such a test is an ambitious project that is well beyond the scope of this Article. In the meantime, however, courts should not sit idle when it

237. 35 U.S.C. § 284 (2006) (emphasis added).

238. *See* Aro Mfg. Co. v. Convertible Top Replacement Co., 377 U.S. 476, 507 (1964) (describing damages analysis as asking, “had the Infringer not infringed, what would the [Patentee] have made?”); *Brooktree Corp. v. Advanced Micro Devices, Inc.*, 977 F.2d 1555, 1579 (Fed. Cir. 1992) (“In patent cases, as in other commercial torts, damages are measured by inquiring: had the tortfeasor not committed the wrong, what would have been the financial position of the person wronged?”).

239. *See supra* Part II.

240. *Cf.* Anna B. Laakman, *An Explicit Policy Lever for Patent Scope*, 19 MICH. TELECOMM. & TECH. L. REV. 43, 77 (2012) (“The Federal Circuit should expressly incorporate contextual factors into its patent scope determinations. This includes the cost of R&D, the ratio of R&D costs to imitation costs, technological risk, and the availability of non-patent alternatives for capturing the social value of inventions.”).

is all too evident that traditional remedies go far afield in certain classes of cases.²⁴¹

Relatedly, the injunctive relief provision, 35 U.S.C. § 283, should be similarly amended. As it stands, it states, “courts . . . may grant injunctions in accordance with the principles of equity to prevent the violation of any right secured by patent, on such terms as the court deems reasonable.”²⁴² Again, this provision should not aim to “prevent the violation of any right secured by patent,” but instead merely to promote optimal innovation incentives. Such a shift would imply that the current four-factor tests for preliminary and permanent injunctions should be suitably modified.²⁴³ Like make-whole damages, the irreparable injury, inadequate compensation, and hardship prongs of the test focus on the private injury to the parties, while only one prong—the public interest factor—concerns society at large. Under a proper model of patent remedies, even if the patentee might experience irreparable harm not curable by monetary damages, which in turn creates a burden that outweighs any experienced by an infringer, an injunction could still be socially detrimental. Ultimately, injunctions should turn only on an analysis that solely concerns the public interest—and not just the interest of the public in using the invention but also its interest in the creation and commercialization of the invention.²⁴⁴

Finally, the notion of willful infringement does not seem to accord well with promoting innovation. In tort law, punitive damages are often awarded for intentional, willful conduct in order to provide greater deterrence of harmful behavior. But infringement of a patent is not harmful *per se*; rather, infringement is only harmful to the extent it denies the patentee an opportunity to be compensated an amount sufficient to induce it

241. John Golden argues that a remedies approach focused on costs would be riddled with measurement difficulties. Golden, *supra* note 65, at 538. Yet, the current approach is also riddled with such difficulties. *See supra* notes 176–180 and accompanying text. Thus, the proper determination is a comparative one, and while most cases today will not justify deviating from the baseline, many will. *See supra* subpart III(C).

242. 35 U.S.C. § 283.

243. *See eBay Inc. v. MercExchange, L.L.C.*, 547 U.S. 388, 391 (2006) (holding that injunctive relief is appropriate for a patentee when the following factors weigh in its favor: “(1) that it has suffered an irreparable injury; (2) that remedies available at law, such as monetary damages, are inadequate to compensate for that injury; (3) that, considering the balance of hardships between the plaintiff and defendant, a remedy in equity is warranted; and (4) that the public interest would not be disserved by a permanent injunction”); *PHG Techs., LLC v. St. John Cos.*, 469 F.3d 1361, 1365 (Fed. Cir. 2006) (noting that the factors for a preliminary injunction are “(1) the likelihood of the patentee’s success on the merits; (2) irreparable harm if the injunction is not granted; (3) the balance of hardships between the parties; and (4) the public interest”).

244. Of course, the “public interest” certainly should not presume that granting injunctions promotes innovation, as the Federal Circuit has been wont to do—even after *eBay*. *See, e.g., Celsis In Vitro, Inc. v. CellzDirect, Inc.*, 664 F.3d 922, 931 (Fed. Cir. 2012) (“We have long acknowledged the importance of the patent system in encouraging innovation.” (quoting *Sanofi–Synthelabo v. Apotex, Inc.*, 470 F.3d 1368, 1383 (Fed. Cir. 2006))).

to engage in innovative activity.²⁴⁵ As I explained earlier, in many situations, some level of infringement will promote overall welfare.²⁴⁶ In other situations, as long as the patentee can recover suitable damages in a lawsuit, whether the infringement was willful becomes a meaningless investigation.

Instead, willful infringement should only play a role in determining damages when it (1) increases the difficulty with which the patentee can detect infringement; or (2) substantially raises the costs of the patentee collecting appropriate compensation for its innovative activity, i.e., by creating the need for a costly lawsuit relative to the value of the innovation. By framing infringement in terms of “wrongful” behavior, Congress and the courts ill-advisedly eradicate some third-party activity that would otherwise serve a salutary purpose.²⁴⁷ In this sense, willfulness damages solely inflicted as “punishment” for “wrongful” behavior should be eliminated from patent doctrine. Rather, infringement should be encouraged just up to the level that suboptimally diminishes incentives to innovate.

B. Towards a Regulatory Model of Patent Law Remedies

So far, I have argued that all private law notions—contract, real property, and tort—should play no fundamental role in how we think about patent rights and remedies.²⁴⁸ A more appropriate model is one that draws on what David Rosenberg terms a “public law vision” of the tort system’s aims.²⁴⁹ Instead of viewing patents as conferring a private right on the patentee, under a public law model, a patent confers a kind of private attorney general status on the patentee that allows it to collect payment *on behalf of society*, which is then immediately remitted in full to the patentee solely in order to optimize overall social innovation incentives.²⁵⁰ Indeed, on a public law model, one could imagine the government bringing patent infringement actions, either in place of or alongside private patentees, then

245. See *supra* section II(C)(2).

246. See *supra* section II(C)(2).

247. See *supra* section II(C)(2).

248. In making this claim, I do not dispute the view that historically the patent system has rested on private law framework. See Kerr, *supra* note 81 (setting forth a “private law theory” of patent law); cf. John M. Golden, *Patent Privateers: Private Enforcement’s Historical Survivors*, 26 HARV. J.L. & TECH. 545, 547–50 (2013) (proposing a “patent privateering” model of enforcement whereby “patents . . . are devices to harness private law enforcement to advance public policy,” namely the constitutionally sanctioned goal of “promot[ing] the Progress of Science and useful Arts.”). Rather, my point is a normative one.

249. See generally David Rosenberg, *The Causal Connection in Mass Exposure Cases: A “Public Law” Vision of the Tort System*, 97 HARV. L. REV. 849 (1984) (positing a “public law” approach to tort law grounded in regulatory aims).

250. Cf. Golden, *supra* note 248, at 612 (limiting the “private attorney general” notion in patent law to validity challenges); La Belle, *supra* note 19, at 50–51 (proposing a “private attorney general” public law model of patent validity challenges but not patent litigation as a whole).

remitting damages to the patentee.²⁵¹ However, because the patentee typically plays an active role in providing factual evidence in any given case—as well as enjoys the direct benefits of a judgment—arguably it is the least-cost enforcer of patent rights. As such, a patentholder will generally be more effective than the government (or some third party for that matter) to collect damages on society's behalf. But, importantly, the choice of patentee as money collector does not stem from any private right of the patentee to be compensated for harm caused to it.²⁵²

In this regard, Shubha Ghosh has offered an insightful public regulatory model of patent law. In Ghosh's view, the patent system should not rest on a social contract or private property foundation but rather "a system of regulation."²⁵³ In other words, "patents serve as a way to administer and organize markets."²⁵⁴ Yet, Ghosh draws particular conclusions about the structure of the patent system that do not in my view necessarily follow from a regulatory model. Specifically, he contends that a regulatory approach counsels in favor of viewing intellectual property law as involving an "assurance game"—namely, one in which "each player independently can be assured that the other[s] will not imitate."²⁵⁵

However, there is no a priori reason why a regulatory model mandates any different game theoretic view of patent law than a private law model. The underlying economic problems associated with patent law are the same regardless of the lens through which we view the rights and remedies designed to solve the problems. Rather, the aim of the regulatory approach should be more general: to optimize incentives for invention and commercialization, taking into account potential costs from diminished

251. See William Kingston, *The "Thesis" Chapters*, in DIRECT PROTECTION OF INNOVATION, xi, 20–27 (William Kingston ed., 1987) (suggesting that the government bring infringement actions in certain classes of cases).

252. Of course, we may wish to incorporate principles of substantive fairness and distributive justice into a public law system—for instance, to protect independent inventors whose life's work is appropriated—but conferring traditional private rights is unnecessary to adequately promote these aims. See LOUIS KAPLOW & STEVEN SHAVELL, FAIRNESS VERSUS WELFARE 43–44 (2002) (discussing how deontological "fairness" constraints can impose limits on otherwise purely consequentialist systems).

253. Ghosh, *supra* note 54, at 1387. Lemley rejects a public regulatory model on the grounds that "[r]egulation is out of vogue, and those who talk about intellectual property as regulation usually do so in order to denigrate it." Lemley, *supra* note 25, at 1074. Of course, this is not a substantive rejection but merely one of form. Here, I do not propose a regulatory model as a means of political reform but as the proper theoretical lens from which to view intellectual property.

254. Ghosh, *supra* note 54, at 1350; see also LANDES & POSNER, *supra* note 109, at 35–36 (addressing the role of the state in regulating intellectual property law).

255. See Ghosh, *supra* note 54, at 1335–36.

follow-on invention, duplicated development and rent-seeking, and consumer deadweight losses, as well as distributive interests.²⁵⁶

Conclusion

The fundamental premise of patent law remedies derives from tort law and counsels that courts should always attempt to make the patentee whole in the event of infringement. Here, I have rejected this premise, arguing that the more fundamental goal of promoting innovation sometimes requires deviations from make-whole remedies. Specifically, when a patent covers a mere component of a patented product, and an infringer's switching costs are high, it will often be more desirable to deny injunctive relief and reduce make-whole damages, regardless of whether the patentee practices its patents. Additionally, an injunction may thwart optimal innovation incentives when it generates large consumer deadweight losses, results in substantial duplicated costs during the pre-invention R&D process, or creates transaction costs far in excess of the value of the invention. Specifically, in these situations, some infringement may be efficient. In other situations, such as when infringement is difficult to detect, increasing damages beyond the make-whole level may be optimal.

More generally, I have argued that the notion that patentees should be returned to the *status quo ante* is a private law concept that should play no essential role in determining optimal remedies in patent law. On this ground, the statutory remedies provisions of the Patent Act rest on a flawed foundation. Instead of correcting for "wrongs" inflicted on private parties, patent law remedies should be tailored to optimize innovation incentives, taking into account the various social costs of patenting. Such a public law vision of the patent system is essentially regulatory in nature. By fully removing the private law moorings of patent law remedies, patents will better fulfill their fundamental role in promoting innovation.

256. Indeed, Ghosh recognizes that "[f]rom the perspective of patent law, the relevant interests include pioneer inventors, follow-on inventors, marketers, consumers, and the innumerable other interests reflected in the research and development processes." *Id.* at 1355.